

0 3

Figure 5 shows a parallel plate capacitor connected to a 3.0 kV dc supply. The capacitor is made from two identical metal plates separated by 5.0 mm of air. Each plate measures 30 cm in length and 20 cm in width.

Figure 5

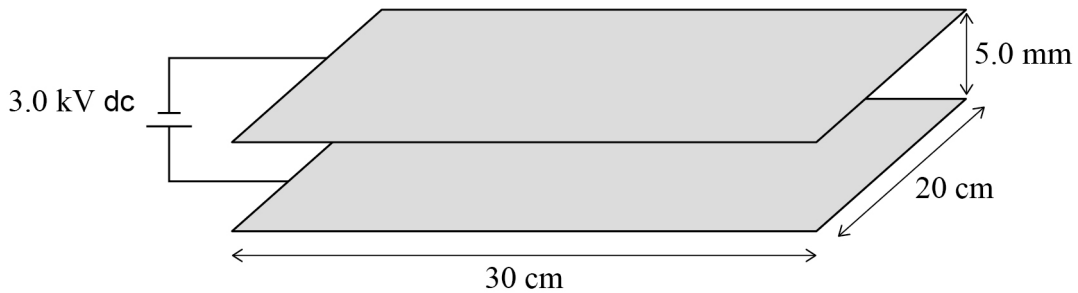


Figure 6 shows a side view of the capacitor.

Figure 6



0 3

1

Draw on **Figure 6** the electric field lines between the plates.

[2 marks]

0 3

2

Calculate the electric field strength between the plates.

[2 marks]

electric field strength = _____ N C^{-1}

Question 3 continues on the next page

Turn over ►



0 3 . 3 The magnitude of the charge stored on each plate is Q .

Show that the electric field strength E can be expressed as

$$E = \frac{Q}{\epsilon_0 \epsilon_r A}$$

where ϵ_r is the relative permittivity of air and A is the area of overlap of the plates.

[2 marks]

0 3 . 4 Calculate Q .

$$\epsilon_r = 1.00$$

[2 marks]

$$Q = \underline{\hspace{4cm}} \text{ C}$$



0 3 . 5 State the electrical properties of a dielectric material.

[2 marks]

0 3 . 6 The capacitor is isolated from the dc supply and a dielectric material is inserted between the plates. The potential difference between the plates changes.

Explain the change in the potential difference.

[2 marks]

12

Turn over for the next question

Turn over ►



Do not write Do not write

A ball of mass 0.2 kg moves in a horizontal circle of radius 50 cm at the end of a string. The ball moves at a constant speed of 9.0 m s^{-1} .

What is the horizontal force exerted by the ball on the string?

[1 mark]

A 8 N away from the centre of the circle

☐

B 8 N towards the centre of the circle

☐

C 32 N away from the centre of the circle

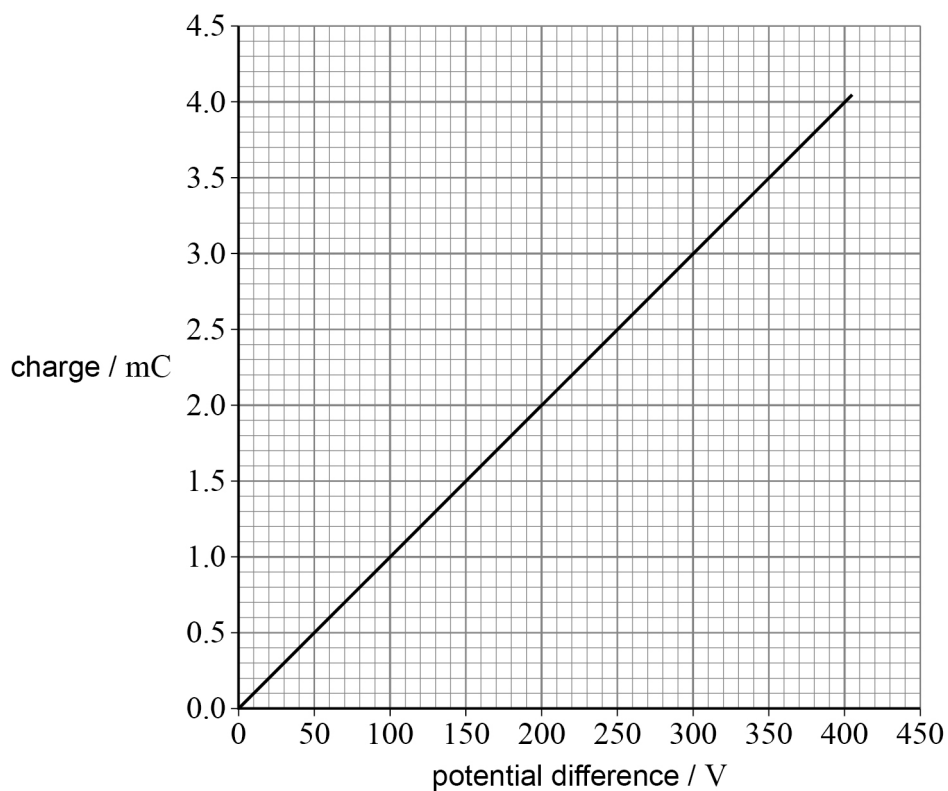
☐

D 32 N towards the centre of the circle

☐

Don't write

The graph shows how the charge on a capacitor varies with the potential difference across the capacitor.



What is the additional energy stored by the capacitor when the potential difference increases from 200 V to 400 V?

[1 mark]

A 0.2 J

☐

B 0.4 J

☐

C 0.6 J

☐

D 0.8 J

☐

Do not write

A charged capacitor of capacitance $460\ \mu\text{F}$ discharges through a fixed resistor. After $184\ \text{s}$ the capacitor has lost 75% of its initial charge.

What is the resistance of the fixed resistor?

[1 mark]

A $2.0 \times 10^5\ \Omega$

☐

B $2.9 \times 10^5\ \Omega$

☐

C $1.4 \times 10^6\ \Omega$

☐

D $2.9 \times 10^8\ \Omega$

☐

Turn over for the next question

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ANSWER IN THE SPACES PROVIDED**

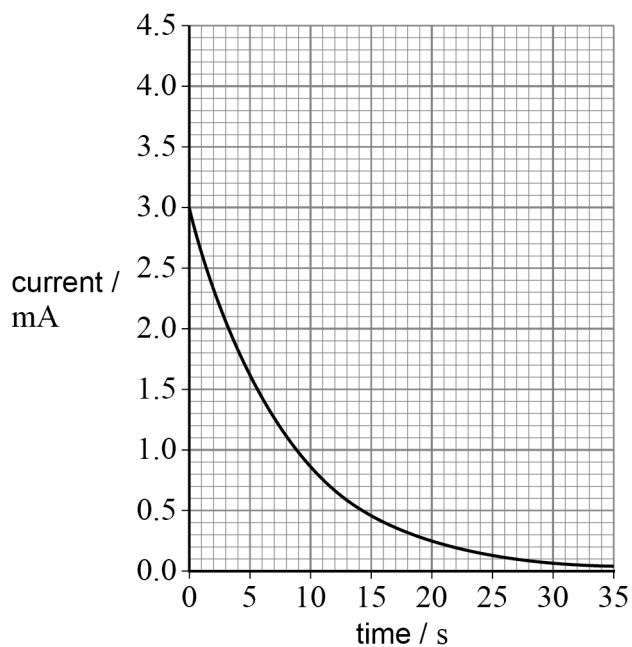
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Do not write Do not write

2 2 A capacitor with an initial potential difference V_0 was discharged through a resistor.

The graph shows the variation of current in the circuit with time.



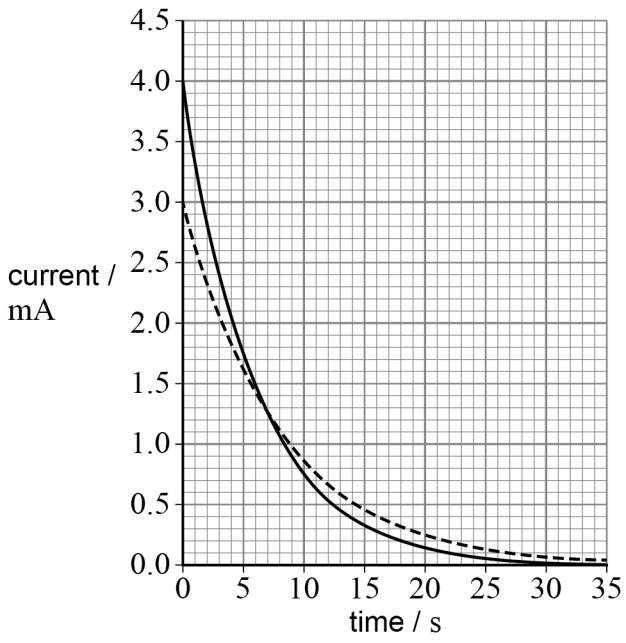
The capacitor was again discharged from V_0 through a second resistor of smaller resistance.

Which graph shows the variation of current with time for the second resistor?
The dashed line shows the variation for the first resistor.

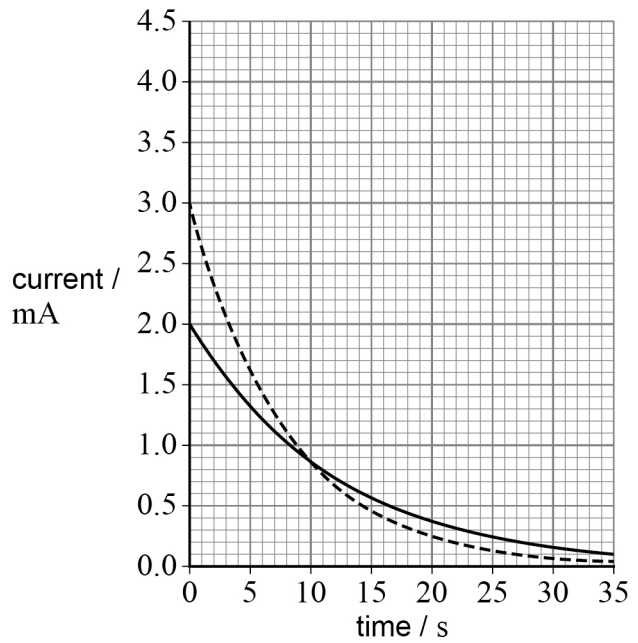
[1 mark]



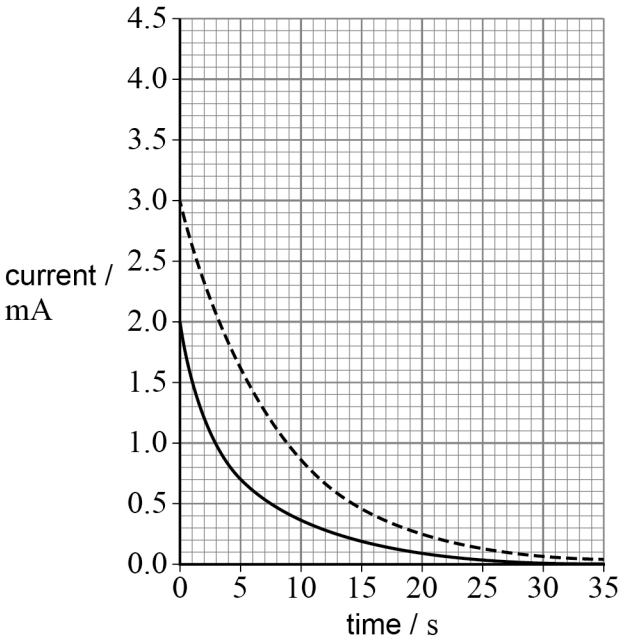
A



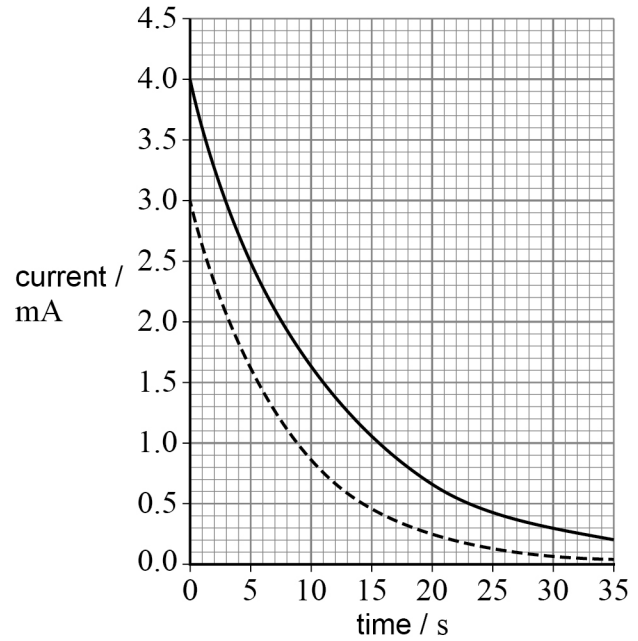
B



C



D



A



B



C



D



Do not write Do not write

2

3

A radioactive source contains a nuclide that emits alpha particles. The energy transformed in each decay is 5.5 MeV.

The source initially produces 60 μW of power. The nuclide has a half-life of 90 years.

What is the activity of the source after 270 years?

[1 mark]

A $8.5 \times 10^6 \text{ Bq}$

☐

B $1.7 \times 10^7 \text{ Bq}$

☐

C $2.3 \times 10^7 \text{ Bq}$

☐

D $8.5 \times 10^9 \text{ Bq}$

☐

3 **2** The frequency of the ac source is 200 Hz.

Which time-base setting on the oscilloscope will display **only two** complete waveforms on the screen?

[1 mark]

A 1 ms cm^{-1}



B 2 ms cm^{-1}



C 5 ms cm^{-1}



D 10 ms cm^{-1}



Don't write

3**3**

The root mean square voltage of the ac source is 40 V.

Which volts / division setting on the oscilloscope will display the largest complete waveform?

[1 mark]

A 5 volts / division



B 10 volts / division



C 20 volts / division



D 50 volts / division



Figure 2 shows a capacitor and a switch **S** connected to a parallel combination of two resistors, **R**₁ and **R**₂. The initial potential difference across the capacitor is 5.0 V and the initial charge, in C, on the capacitor is Q_0 .

Figure 2

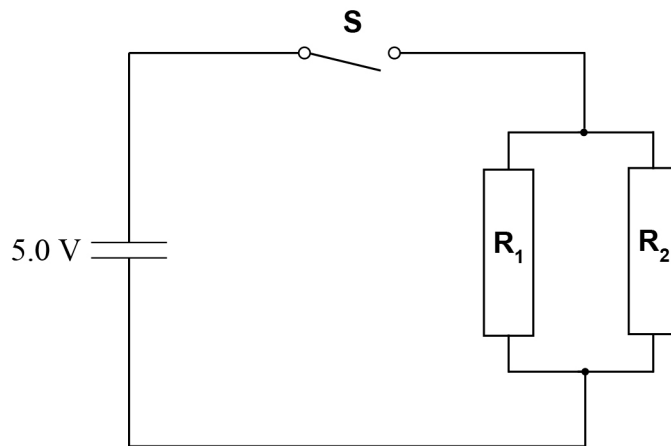
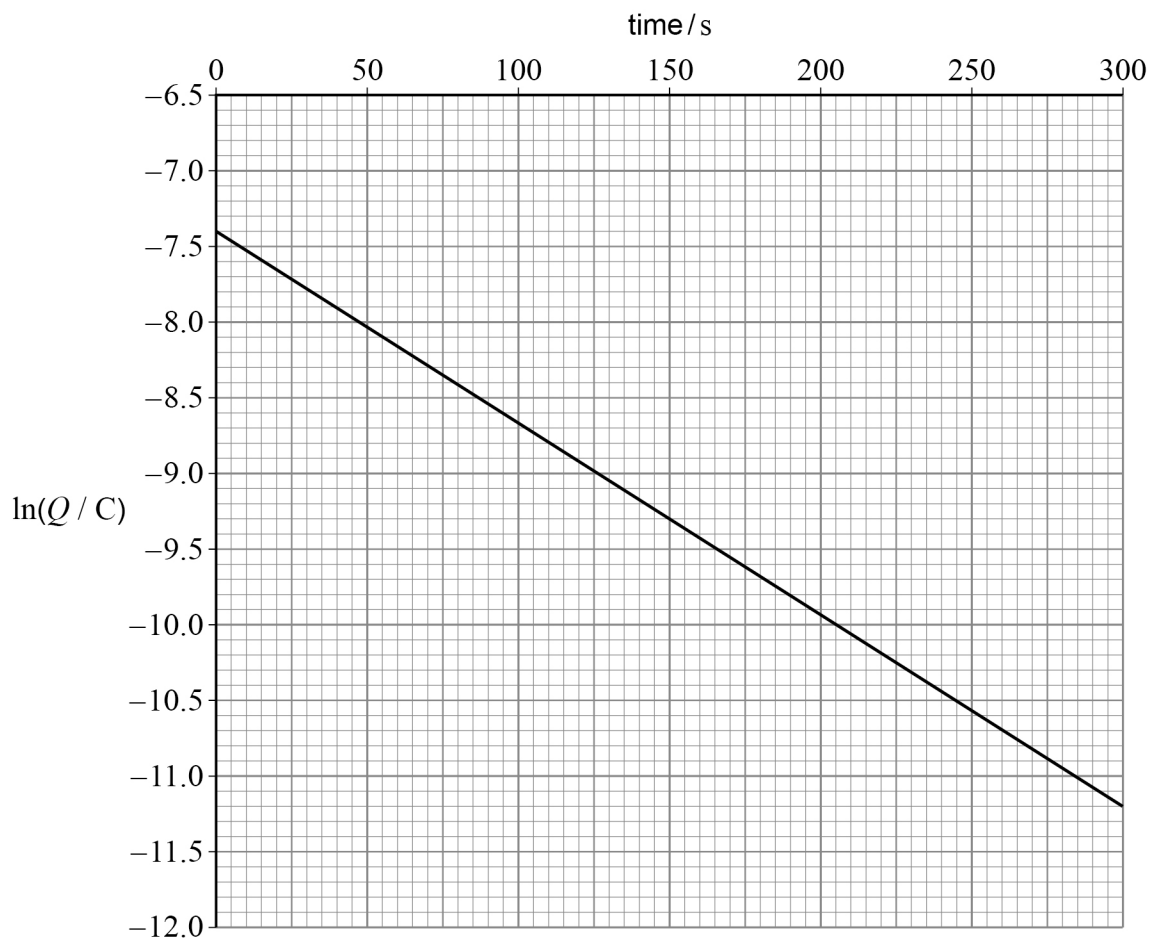


Figure 3 shows the variation of $\ln(Q / C)$ with time after **S** is closed.

Figure 3



0 2 . 1 Determine Q_0 .

[2 marks]

$$Q_0 = \underline{\hspace{4cm}} \text{ C}$$

0 2 . 2 Show that the capacitance of the capacitor is approximately $120 \mu\text{F}$.

[1 mark]

0 2 . 3 Show, using **Figure 3**, that the time constant of the circuit is approximately 80 s .

[3 marks]

Question 2 continues on the next page

Turn over ►



0 2 . 4 The resistance of R_1 is $3.2 \text{ M}\Omega$.

Calculate, in $\text{M}\Omega$, the resistance of R_2 .

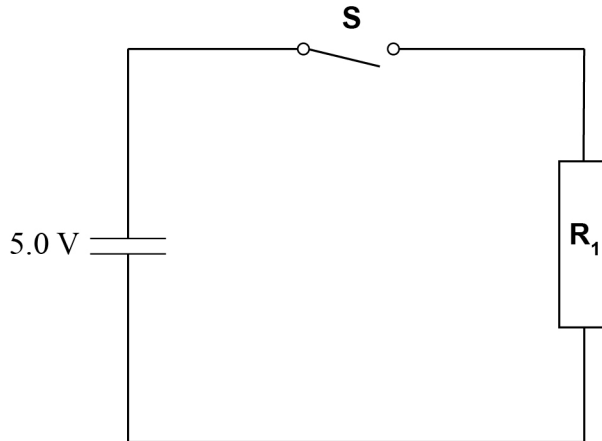
[3 marks]

resistance of $R_2 =$ _____ $\text{M}\Omega$



0 2 . 5 **Figure 4** shows the circuit with **S** open and **R**₂ removed.

Figure 4



The capacitor is recharged so that the initial potential difference across the capacitor is 5.0 V.

S is now closed.

Compare the initial rate of discharge of the capacitor in **Figure 4** with the initial rate of discharge of the capacitor in **Figure 3**.

[3 marks]

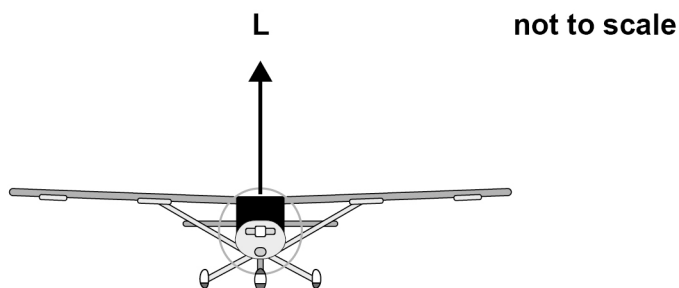


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03

Figure 5 shows an aircraft flying at a constant height. A lift force **L** acts at right angles to the aircraft's wings. The magnitude of **L** depends on the speed of the aircraft.

Figure 5



The mass of the aircraft is 1100 kg.

03

. 1

Calculate the magnitude of **L**.

[1 mark]

magnitude of **L** = _____ N

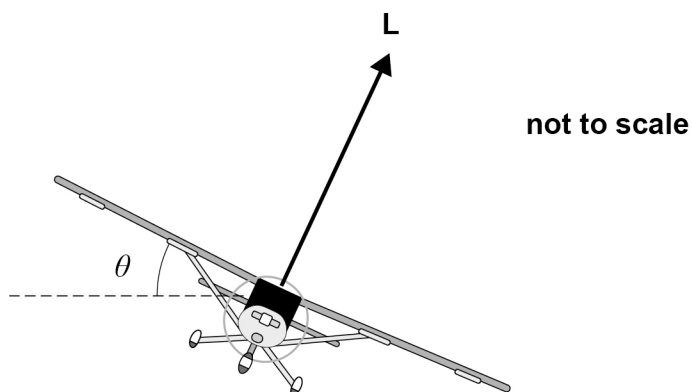


0 3 . 2

Figure 6 shows the aircraft travelling at an increased speed with its wings making an angle θ to the horizontal.

The magnitude of **L** has increased. The aircraft moves in a horizontal circle at a constant speed.

Figure 6



A resultant force **R** acts on the aircraft in **Figure 6**. **R** is produced by the weight **W** of the aircraft and **L**.

Draw, in the space below, a vector diagram to show how the addition of **W** and **L** produces **R**.

[2 marks]

Question 3 continues on the next page

Turn over ►



0 3 . 3 In **Figure 6**, $\theta = 25^\circ$.

Show that **L** is approximately 12 kN.

[2 marks]

0 3 . 4 When $\theta = 25^\circ$, the radius of the circle is 900 m.

Calculate the speed of the aircraft.

[3 marks]

speed = _____ m s^{-1}



0 3 . 5

The angle θ is increased to more than 25° . The speed of the aircraft does not change from the speed in Question **03.4**.

State and explain the changes in the motion of the aircraft.

[4 marks]

12

Turn over for the next question

Turn over ►



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1	1
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Two apples are touching each other.

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What is the best estimate for the gravitational force between the two apples?

[1 mark]

A 10^{-20} N

☐

B 10^{-15} N

☐

C 10^{-10} N

☐

D 10^{-5} N

☐

1 6

A deuterium atom consists of one electron and a nucleus.

The nucleus has one proton and one neutron.

F_E is the electrostatic force on the electron due to the deuterium nucleus.

F_G is the gravitational force on the electron due to the deuterium nucleus.

What is the ratio $\frac{F_E}{F_G}$?

[1 mark]

A 8.4×10^{18}

☐

B 1.2×10^{36}

☐

C 1.1×10^{39}

☐

D 2.3×10^{39}

☐

Turn over for the next question

Turn over ►



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What is the unit of capacitance in fundamental (base) SI units?

[1 mark]

A $\text{A}^2 \text{kg}^{-1} \text{m}^{-2}$

☐

B $\text{A}^2 \text{s}^4 \text{kg}^{-1} \text{m}^{-2}$

☐

C $\text{A}^2 \text{s}^5 \text{kg}^{-1} \text{m}^{-2}$

☐

D $\text{C}^2 \text{s}^2 \text{kg}^{-1} \text{m}^{-2}$

☐

Don't waste

2

2

A parallel plate capacitor is charged and isolated.

A dielectric material is inserted between the two plates of the capacitor.

What happens to the potential difference across the capacitor and the energy stored by the capacitor when the dielectric is inserted?

[1 mark]

	Potential difference	Energy	
A	decreases	decreases	☐
B	decreases	increases	☐
C	increases	decreases	☐
D	increases	increases	☐

2 3

A capacitor has capacitance 180 mF .

The capacitor is charged by a constant current for 36 s .

During this time, the potential difference across the capacitor increases from 3.0 V to 9.0 V .

What is the magnitude of the constant current?

[1 mark]

A 5 mA

☐

B 30 mA

☐

C 45 mA

☐

D 60 mA

☐

Turn over ►



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2**4**Do not write
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A capacitor of capacitance $100\ \mu\text{F}$ stores charge Q_0 .

The capacitor is discharged through a $120\ \text{k}\Omega$ fixed resistor.

What is the time taken for the charge on the capacitor to become $0.25Q_0$?

[1 mark]

A 4.2 s

☐

B 8.3 s

☐

C 17 s

☐

D 24 s

☐

2**5**

A $220\ \mu\text{F}$ capacitor is charged to a potential difference of $6.0\ \text{V}$. The capacitor is then discharged through an $18\ \text{k}\Omega$ resistor.

What is the potential difference across the resistor after $2.0\ \text{s}$?

[1 mark]

A $0.6\ \text{V}$

☐

B $2.4\ \text{V}$

☐

C $3.6\ \text{V}$

☐

D $4.4\ \text{V}$

☐

An ac power source has a peak power of P_0 and a root mean power of P_{rms} .

What is $\frac{P_0}{P_{\text{rms}}}$?

[1 mark]

A $\frac{1}{2}$

☐

B $\frac{1}{\sqrt{2}}$

☐

C $\sqrt{2}$

☐

D 2

☐

Turn over for the next question

Turn over ►

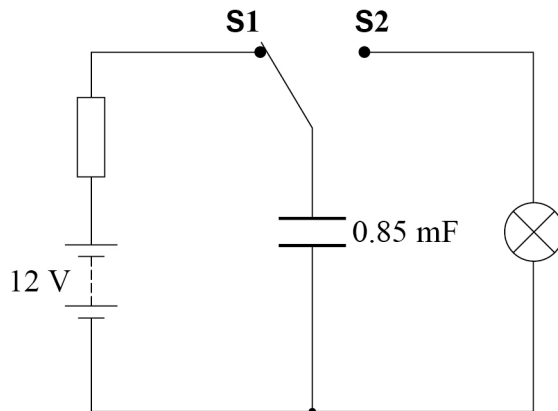


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0 6

Figure 6 shows a circuit used to transfer energy to the lamp in a camera flash.

Figure 6



A battery of emf 12 V and negligible internal resistance charges the capacitor when the switch is in position **S1**.

The capacitor discharges through the lamp when the switch is in position **S2**. Assume that the capacitor is discharged completely after each use.

0 6

1

The capacitor has a capacitance of 0.85 mF.

Show that the energy stored in the capacitor when it is fully charged is approximately 0.06 J.

[1 mark]

0 6

2

The lamp emits a flash of visible light when the voltage across it is greater than a minimum voltage $V_{\min}$.

The duration of the flash is 55 ms.

Assume that the resistance of the lamp is constant and has a value of 31 Ω .

Calculate $V_{\min}$.

[2 marks]

$V_{\min} =$ _____ V



The charging circuit includes a resistor and a 12 V battery. The battery transfers a total energy of 50 mW h during its lifetime.

0 6 . 3

Explain why only half of this energy can be transferred to the capacitor.

[2 marks]

0 6 . 4

Calculate the maximum number of times the battery can charge the capacitor during the battery's lifetime.

[2 marks]

maximum number of times = _____

Question 6 continues on the next page



The capacitor is replaced with one of much greater capacitance.

0 6 . 5

Explain how the replacement capacitor affects the duration and the brightness of each flash.

[3 marks]

0 6 . 6

Explain how the replacement capacitor affects the number of flashes the battery can deliver.

Assume that the capacitor discharges completely each time the flash is used.

[2 marks]



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional sheets for this working.

An object moves at a constant speed in a horizontal circle.

Which statement is true?

[1 mark]

A The object is in equilibrium.

☐

B The velocity of the object is constant.

☐

C The total energy of the object is constant.

☐

D There is a resultant force on the object in the direction of its motion.

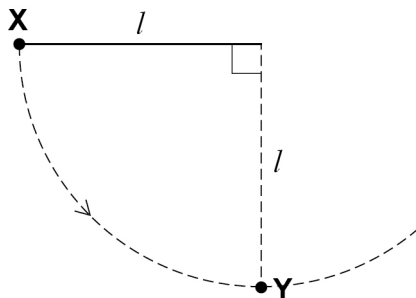
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An object of mass m on the end of a string of length l is released from rest at **X**. It moves in a circular path and passes through **Y** when the string is vertical.



What is the tension in the string when the object passes through **Y**?

[1 mark]

A mg

☐

B $3mg$

☐

C $mg l$

☐

D $\sqrt{2gl}$

☐

1

1

The separation between the Earth and the Sun varies.

The minimum separation is 1.7% smaller than the average separation.

The maximum separation is 1.7% greater than the average separation.

What is $\frac{\text{force of attraction between Earth and Sun at the minimum separation}}{\text{force of attraction between Earth and Sun at the maximum separation}}$?

[1 mark]

A 0.934



B 0.967



C 1.035



D 1.070

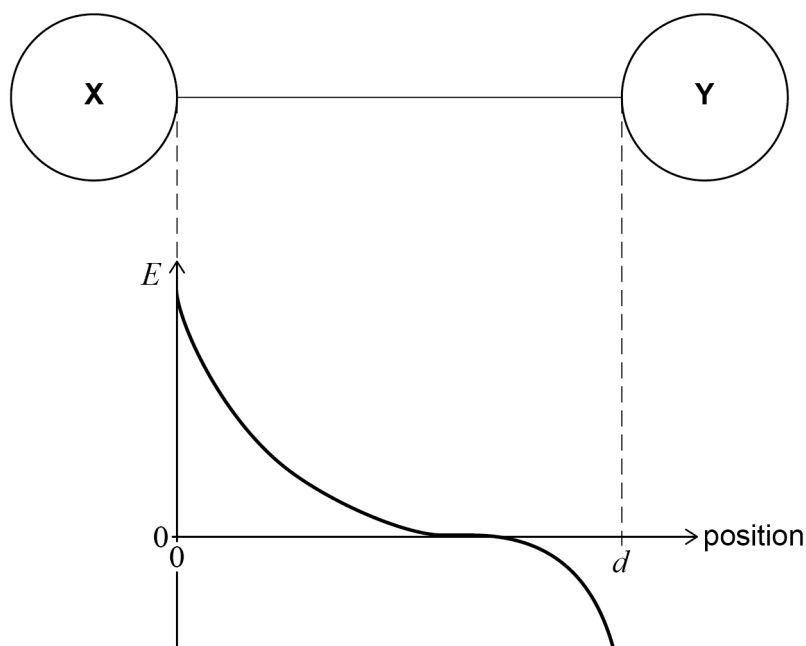


1 5

Two charged bodies **X** and **Y** have the same size, and their surfaces are separated by a distance d .

The charge on **X** is Q_X and the charge on **Y** is Q_Y .

The graph shows the variation of the resultant electric field strength E with position along the line joining **X** and **Y**.



What are the relationships between the signs of Q_X and Q_Y and between the magnitudes of Q_X and Q_Y ?

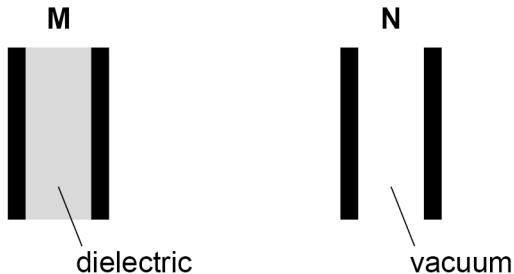
[1 mark]

	Signs	Magnitudes	
A	Q_X and Q_Y have opposite signs	The magnitude of Q_Y is greater than the magnitude of Q_X	☐
B	Q_X and Q_Y have opposite signs	The magnitude of Q_X is greater than the magnitude of Q_Y	☐
C	Q_X and Q_Y have the same sign	The magnitude of Q_Y is greater than the magnitude of Q_X	☐
D	Q_X and Q_Y have the same sign	The magnitude of Q_X is greater than the magnitude of Q_Y	☐



Questions **16** and **17** are about two parallel-plate capacitors **M** and **N**.

M and **N** have identical dimensions.



In **M**, the plates are separated by a dielectric with a relative permittivity of 9

In **N**, the plates are separated by a vacuum.

M and **N** are both charged so that they store the same energy.

What is $\frac{\text{charge stored by M}}{\text{charge stored by N}}$?

[1 mark]

A $\frac{1}{9}$

☐

B $\frac{1}{3}$

☐

C 3

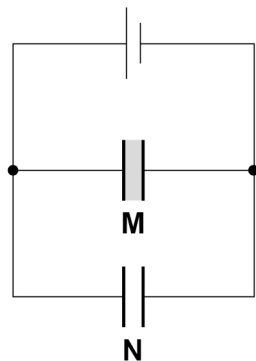
☐

D 9

☐

do not write

M and **N** are discharged and then connected in parallel to a cell.



What is $\frac{\text{charge stored by M}}{\text{charge stored by N}}$ now?

[1 mark]

A $\frac{1}{9}$

B $\frac{1}{3}$

C 3

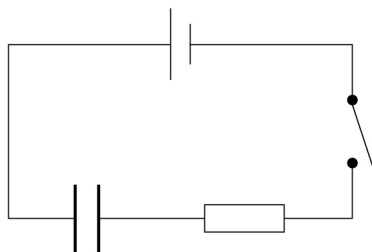
D 9

Turn over for the next question



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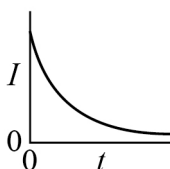
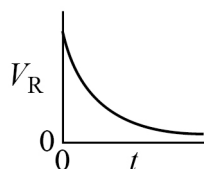
A capacitor is initially uncharged. The switch is closed at time $t = 0$



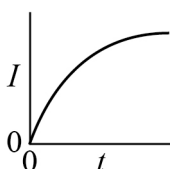
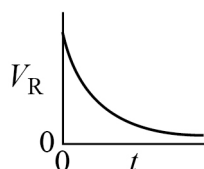
Which pair of graphs shows the variation with time t of the voltage V_R across the resistor and the current I in the circuit after the switch is closed?

[1 mark]

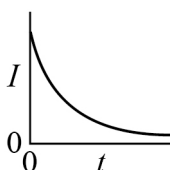
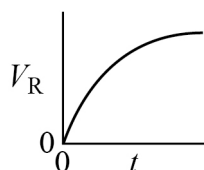
A



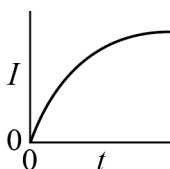
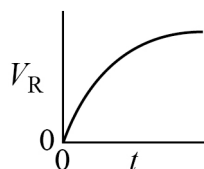
B



C



D



do not write

Figure 6 shows two charged water droplets **A** and **B** in a thundercloud.

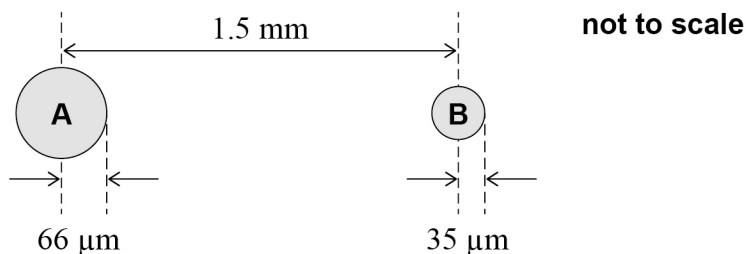
The radius of **A** is $66\ \mu\text{m}$ and the radius of **B** is $35\ \mu\text{m}$.

Each droplet has a charge of $-1.2 \times 10^{-12}\ \text{C}$.

Assume that the charge on **A** and the charge on **B** are concentrated at their centres.

The distance between the centres of **A** and **B** is $1.5\ \text{mm}$.

Figure 6



04.1

Calculate the electrostatic force between **A** and **B**.

[2 marks]

force = _____ N

04.2

Movement of the air in the cloud causes **A** and **B** to move together until they touch.

Calculate the work done against the electrostatic force in moving **A** and **B** from the position shown in **Figure 6** to the position where they touch.

[3 marks]

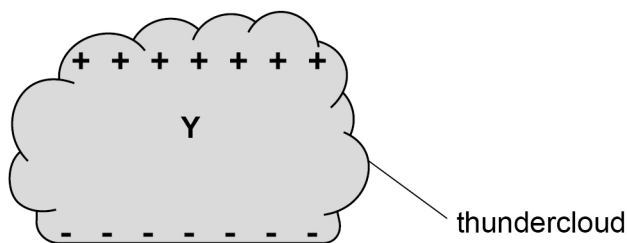
work done = _____ J

Question 4 continues on the next page

Turn over ►

Figure 7 shows a simplified diagram of a thundercloud. The bottom of the thundercloud is negatively charged and the top is positively charged.
Y is a point inside the thundercloud.

Figure 7



E_Y is the electric field strength at **Y**.

0 4 . 3

State the direction of E_Y .

[1 mark]

0 4 . 4

A charged water droplet is stationary at **Y**.
The weight of the droplet and the force on it due to E_Y are equal in magnitude.
The mass of the droplet is 1.38×10^{-9} kg.
The charge on the droplet is -2.1×10^{-12} C.

Calculate E_Y .

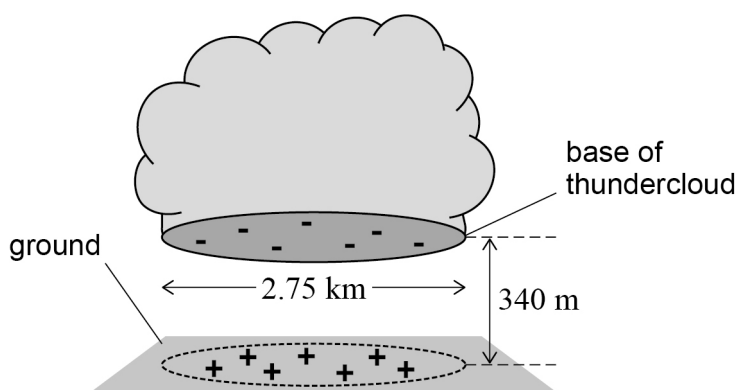
[3 marks]

$E_Y =$ _____ V m^{-1}



The base of the thundercloud and the ground beneath it can be modelled as two plates of a parallel-plate capacitor with air between them as shown in **Figure 8**.

Figure 8



The base of the cloud is circular, with a diameter of 2.75 km.
The base of the cloud is 340 m above the ground.

0 4 . 5

Show that the capacitance of the capacitor is about $1.5 \times 10^{-7} \text{ F}$.

[2 marks]

0 4 . 6

The capacitor discharges when lightning strikes the ground.
The charge stored in the capacitor before it discharges is 87 C.

Calculate the energy available for the lightning strikes.

[2 marks]

energy = _____ J

Do not write Do not write

1**0**

What is the initial path of the particle in the field?

[1 mark]

A circular, curving upwards



B parabolic, curving upwards



C circular, curving downwards



D parabolic, curving downwards



Don't write

1	2
---	---

A capacitor of capacitance C initially stores charge Q_0 .
It is discharged through a resistor of resistance R .

What percentage of Q_0 has been transferred from the capacitor in a time equivalent to $2RC$ after the discharge begins?

[1 mark]

A 14%

☐

B 39%

☐

C 61%

☐

D 86%

☐

Turn over ►

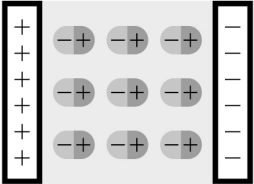
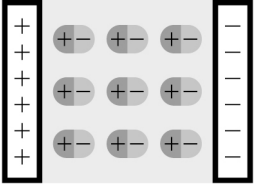
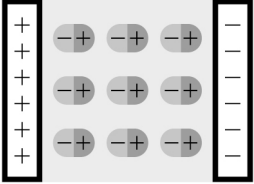
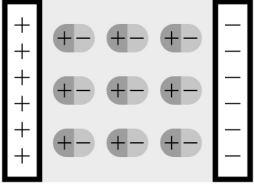


Do not write Do not write

1 5

A dielectric material contains polar molecules that rotate in an electric field. The dielectric material is inserted between the plates of a charged parallel-plate capacitor. The capacitance of the capacitor changes.

Which row shows the alignment of the polar molecules and the effect on the capacitance?
[1 mark]

	Alignment of polar molecules	Effect on capacitance	
A		decreases	☐
B		decreases	☐
C		increases	☐
D		increases	☐

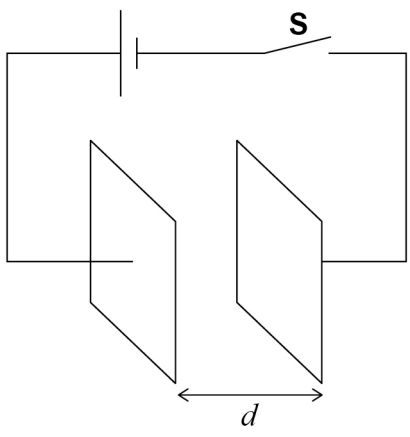
0 5 . 1

State what is meant by a capacitance of 37 mF.

[1 mark]

A capacitor is made up of two parallel metal plates, which are separated in air by a distance d .
A cell of negligible internal resistance is connected across the capacitor as shown in **Figure 10**. Switch **S** is closed and the capacitor becomes fully charged.

Figure 10



0 5 . 2

While **S** is still closed, the plate separation is reduced to $\frac{d}{2}$.

State and explain what happens to the magnitude of the charge stored by the capacitor.

[3 marks]

Question 5 continues on the next page

Turn over ►



0 5 . 3 Explain how charge is conserved when the plate separation is reduced.

[1 mark]

0 5 . 4 **S** is now opened.
The plate separation is increased to d .

State and explain how the energy stored in the capacitor changes during this process.

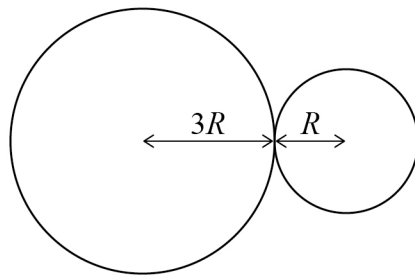
[3 marks]

0 5 . 5 Explain how energy is conserved when the plate separation is increased as described in Question **05.4**.

[1 mark]

Don't write

Two uniform solid spheres of radius R and $3R$ touch. They have the same density ρ .



What is the gravitational force between them?

[1 mark]

A $\frac{5}{6}\pi GR\rho^2$

☐

B $\pi^2 GR^2 \rho^2$

☐

C $3\pi^2 GR^4 \rho^2$

☐

D $\frac{4}{9}\pi^2 GR^5 \rho^2$

☐

1 3

The average mass of a nucleon in a nucleus of the atom A_ZX is M_n .

The mass of an electron is m_e .

What is $\frac{\text{gravitational force}}{\text{electrostatic force}}$ for the forces between the nucleus and an electron in the atom?

[1 mark]

A $\frac{4\pi\epsilon_0 G Z m_e M_n}{A e^2}$

☐

B $\frac{4\pi\epsilon_0 G A m_e M_n}{Z e^2}$

☐

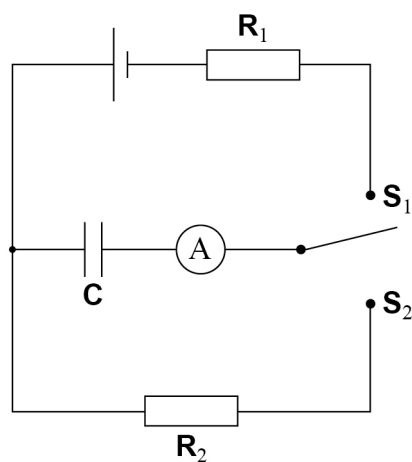
C $\frac{A m_e M_n}{4\pi\epsilon_0 G Z e^2}$

☐

D $\frac{G A m_e M_n}{4\pi\epsilon_0 Z e^2}$

☐

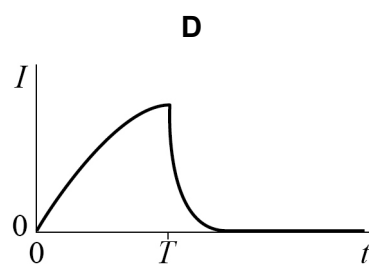
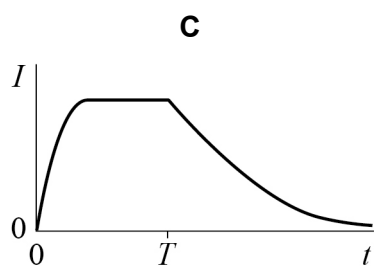
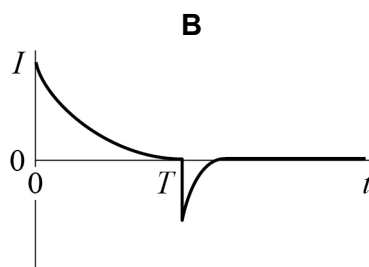
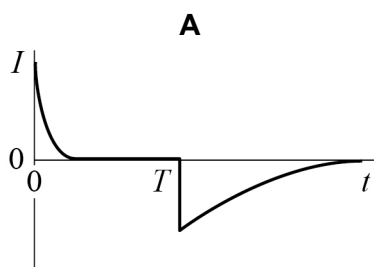
- 1 5** In the circuit, the resistance of resistor R_2 is greater than the resistance of resistor R_1 . The capacitor C is initially uncharged.



At time $t = 0$ the switch is set to position S_1 .
At time $t = T$ the switch is moved to position S_2 .

Which graph shows the variation of current I in the ammeter with t ?

[1 mark]



- A** ☐
- B** ☐
- C** ☐
- D** ☐



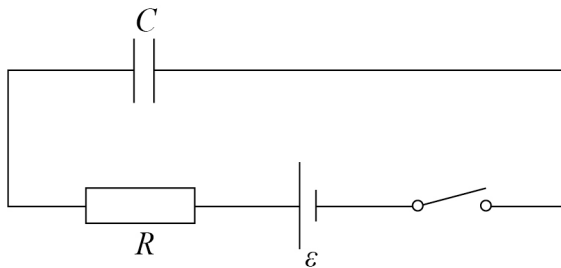
do not write

1 6

A capacitor of capacitance C is connected to a switch, a cell of emf ε and a resistor of resistance R .

The cell has negligible internal resistance.

The capacitor is initially uncharged.



The switch is then closed.

What is the time taken for the voltage across the capacitor to reach $\frac{\varepsilon}{3}$?

[1 mark]

A $RC \ln 3$

☐

B $RC \ln \frac{3}{2}$

☐

C $RC (\ln 2 - \ln 3)$

☐

D $\frac{1}{3} RC \ln 2$

☐

1 9 The resultant force on the electron is zero.

Which equation relates E , B and v ?

[1 mark]

A $Bv = E$

☐

B $B = vE$

☐

C $v = BE$

☐

D $BEv = 1$

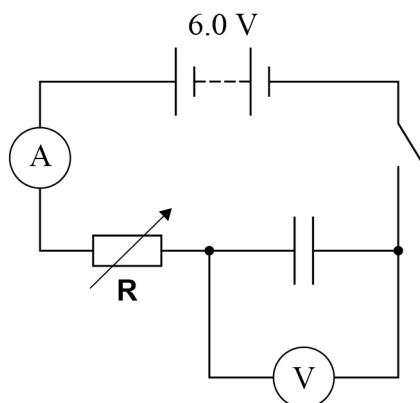
☐

Do not write

A student uses the circuit in **Figure 8** to determine the capacitance of a capacitor. The capacitor is initially uncharged.

The variable resistor **R** is set to its maximum value and the switch is closed. **R** is then used to keep the charging current constant while the capacitor charges.

Figure 8



The charging current is 0.38 mA. During the experiment, the student gradually decreases the resistance of **R** to maintain a constant current of 0.38 mA.

The reading V on the voltmeter is recorded every ten seconds for 120 s. The time is measured using a stopwatch.

The battery has an emf of 6.0 V and negligible internal resistance.

Calculate the maximum resistance of **R**.

[1 mark]

maximum resistance = _____ Ω

Explain why the resistance of **R** needs to be decreased to maintain a constant current.

[1 mark]

Question 4 continues on the next page

Turn over ►

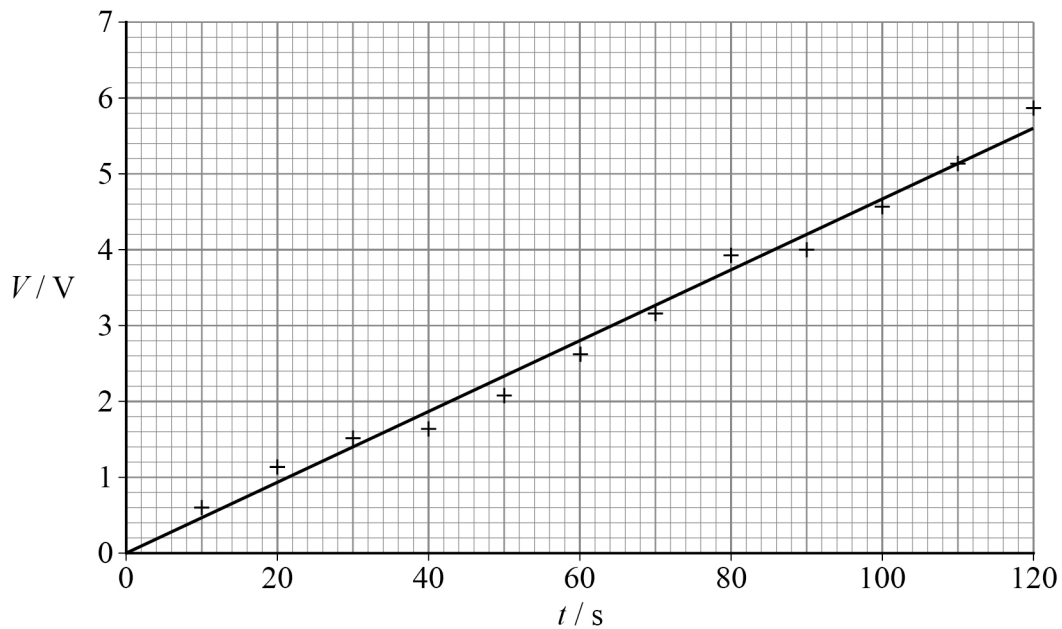


0 4 . 3 Show that the charge on the capacitor after 120 s is approximately 0.046 C.

[1 mark]

Figure 9 is a graph of the voltmeter readings V against time t .

Figure 9



0 4 . 4 Determine the capacitance of the capacitor.

[2 marks]

capacitance = _____ F

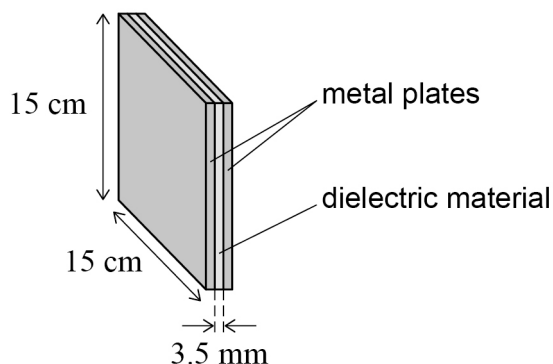


The student makes the new capacitor shown in **Figure 10**.

The two square metal plates have sides of length 15 cm. They overlap completely and are separated by a piece of dielectric material.

The dielectric material has a thickness of 3.5 mm and completely fills the space between the plates.

Figure 10



The capacitance of this capacitor is 0.13 nF.

0 4 . 5

Calculate the relative permittivity of the dielectric material.

[2 marks]

relative permittivity = _____

0 4 . 6

Discuss whether the student could use the experimental procedure described on **page 13** to confirm the capacitance of the 0.13 nF capacitor.

[3 marks]



Do not write

1 3

A parallel-plate capacitor is connected across a battery.
A piece of dielectric material is inserted into the space between the plates with the battery still connected.

What happens to the charge stored on the capacitor and the potential difference across the capacitor?

[1 mark]

	Charge stored on the capacitor	Potential difference across the capacitor	
A	increases	stays the same	☐
B	increases	increases	☐
C	stays the same	stays the same	☐
D	stays the same	decreases	☐

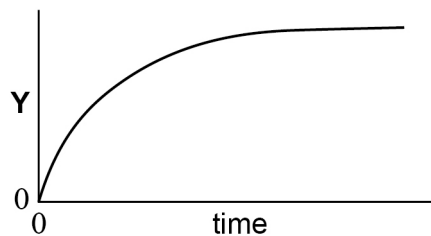
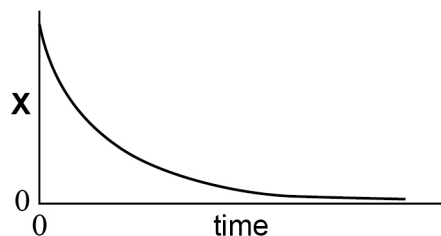
Turn over for the next question



Do not write

1 4

An uncharged capacitor **P** is charged through a resistor **R**.
Quantities **X** and **Y** are measured as the capacitor is charged.
The following graphs are produced.



What are **X** and **Y**?

[1 mark]

	X	Y	
A	potential difference across R	current in R	☐
B	potential difference across P	charge on P	☐
C	potential difference across P	current in R	☐
D	current in R	potential difference across P	☐



*Do not write
outside the
box*

The data below relate to Question **15** and Question **16**.

A 2.2 mF capacitor is charged to a potential difference of 12 V.

It is then discharged through a 10 k Ω resistor until the potential difference across the capacitor decreases to 6.0 V.

1**5**

What is the time taken from the start of the discharge for the potential difference across the capacitor to decrease to 6.0 V?

[1 mark]

A 11 s

☐

B 15 s

☐

C 22 s

☐

D 32 s

☐

How much energy is dissipated in the resistor while the capacitor is discharging from 12 V to 6.0 V?

[1 mark]

A 40 mJ

☐

B 80 mJ

☐

C 120 mJ

☐

D 160 mJ

☐

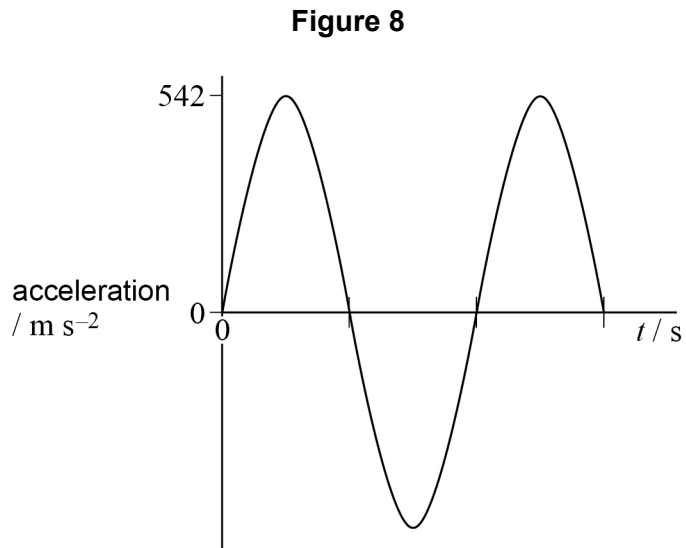
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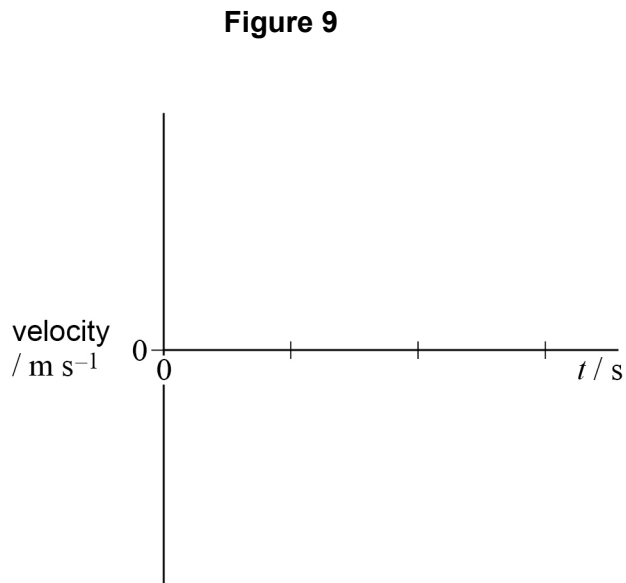
Figure 8 shows the variation of the acceleration of the coil with time t for 1.5 cycles of the motion.



Sketch, on **Figure 9**, the variation of the velocity of the coil with t for 1.5 cycles of the motion.

Show the values of the period of oscillation and the maximum velocity on your sketch.

[2 marks]



0 3 . 1

Show that a geosynchronous satellite orbits the Earth at a distance of approximately 36 000 km above the surface of the Earth.

[4 marks]

0 3 . 2

Show that the orbital speed of the satellite is approximately 3100 m s^{-1} .

[2 marks]



0 3 . 3 The satellite has a mass of 380 kg.

Calculate the total energy of the satellite when it is in a geosynchronous orbit.

[3 marks]

energy = _____ J

0 3 . 4 Geostationary satellites are geosynchronous satellites that remain above the same point of the Earth's surface.

Geostationary satellites are usually launched towards the east from a point on the Equator.

Suggest why this is done.

[3 marks]



0 4

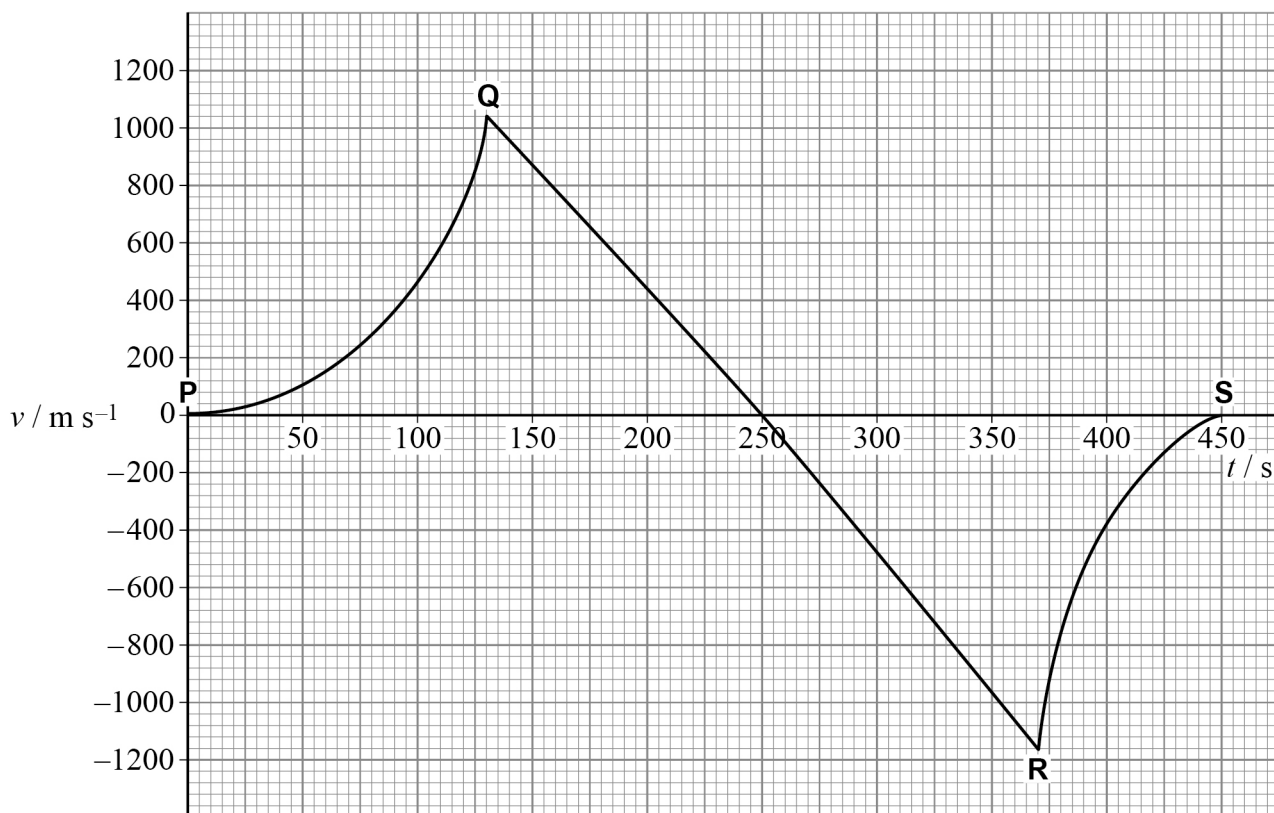
A spacecraft carrying astronauts is launched vertically from the Earth's surface.

The spacecraft's engines stop 130 s after the launch. The spacecraft continues to rise until it reaches its maximum altitude of 106 km. It then falls freely back to Earth.

Parachutes open 370 s after the launch to slow the descent of the spacecraft. The spacecraft returns to the Earth's surface 450 s after the launch.

Figure 10 shows the variation of velocity v with time t for the flight.

Figure 10



0 4 . 1

Calculate the mean magnitude of the acceleration due to gravity for the part of the flight between **Q** and **R**. Air resistance is negligible for this part of the flight.

[2 marks]

mean acceleration due to gravity = _____ m s^{-2}



An astronaut pilots the spacecraft.

0 4 . 2

Describe how the weight of the astronaut varies during the flight.

[1 mark]

During one part of the flight, the astronaut has no sensation of weight.

0 4 . 3

Explain why the astronaut feels weightless during part of the flight.

[1 mark]

0 4 . 4

Describe how the astronaut's sensation of weight varies during the other parts of the flight.

[2 marks]

6

Turn over for the next question

Turn over ►



0	5
---	---

Iodine-124 ($^{124}_{53}\text{I}$) is used in the detection of cancers in the thyroid gland.

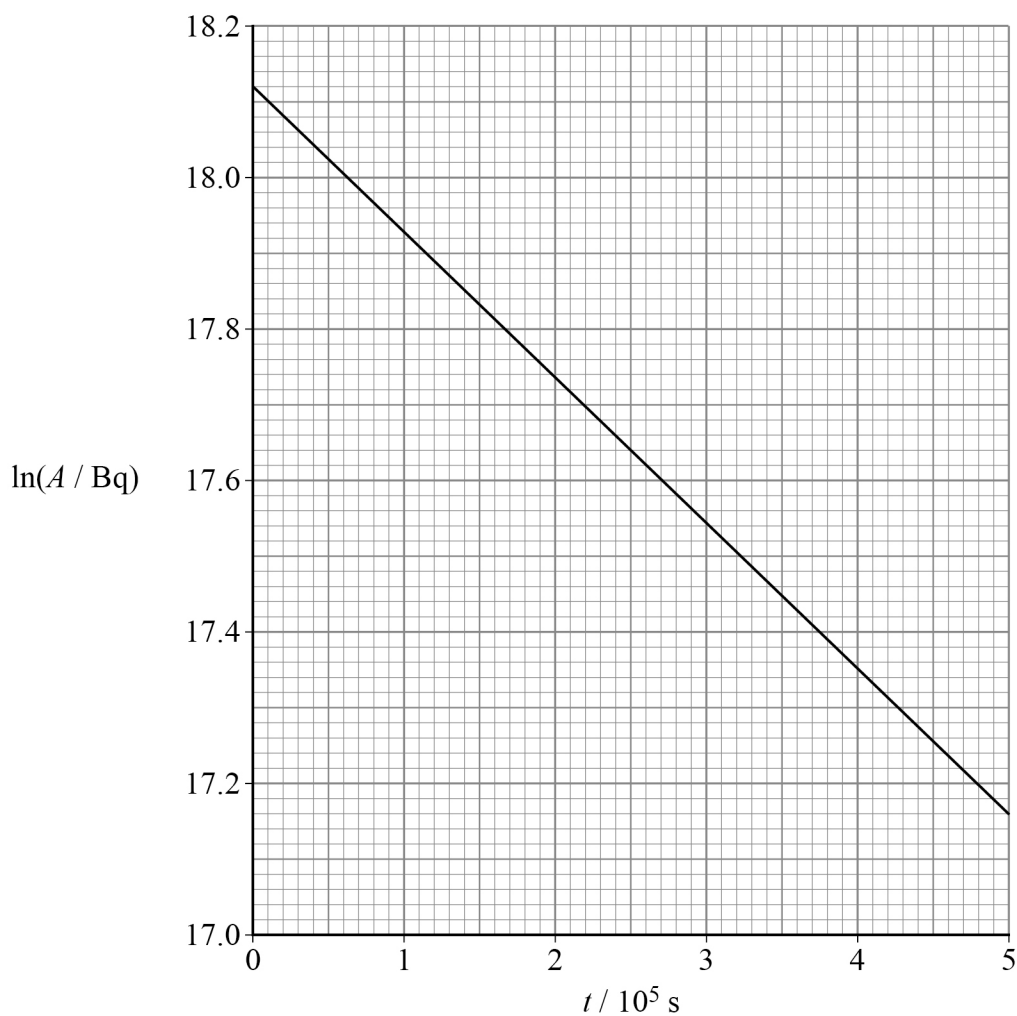
A dose of iodine-124 is given to a patient at time $t = 0$

The iodine-124 is absorbed by the thyroid gland and emissions from decaying iodine-124 are detected outside the patient's body.

The activity A of the dose of iodine-124 decreases with time.

Figure 11 shows the variation of $\ln(A / \text{Bq})$ with time t .

Figure 11



0 5 . 1 Determine the activity of the dose at $t = 0$

[2 marks]

activity = _____ Bq

0 5 . 2 Determine the half-life of iodine-124.

[3 marks]

half-life = _____ s

Question 5 continues on the next page

Turn over ►



0 5 . 3

A different patient is given a dose of iodine-131.

Calculate the mass of iodine-131 in the dose when its activity is 24 MBq.

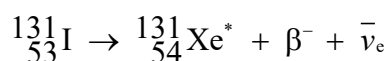
$$\begin{aligned} \text{decay constant of iodine-131} &= 1.0 \times 10^{-6} \text{ s}^{-1} \\ \text{molar mass of iodine-131} &= 0.131 \text{ kg} \end{aligned}$$

[3 marks]

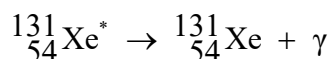
mass = _____ kg

0 5 . 4

$^{131}_{53}\text{I}$ decays into $^{131}_{54}\text{Xe}^*$:



$^{131}_{54}\text{Xe}^*$ then decays by γ emission:



Only one type of particle from this decay series is detected outside the body.

Explain why.

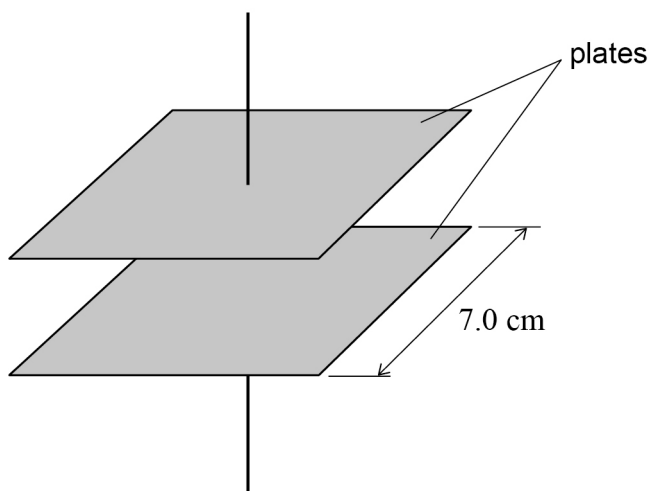
[2 marks]



0 6 . 1

Figure 12 shows a capacitor with two overlapping parallel square plates with sides of length 7.0 cm.

Figure 12



There is an air gap between the plates. The air acts as an insulator when the electric field strength in the gap does not exceed $3.0 \times 10^4 \text{ V m}^{-1}$.

Estimate the maximum charge that can be stored on the capacitor.

[3 marks]

maximum charge = _____ C

Question 6 continues on the next page

Turn over ►



0 6 . 2

A parallel-plate capacitor is charged.

When a dielectric material is inserted between the plates, the capacitance of the capacitor increases.

Explain this increase in capacitance.

[2 marks]



Supercapacitors have very high capacitances. Engineers think that supercapacitors may replace batteries in some applications.

Supercapacitor **X** has a capacitance of 144 F and has a potential difference of 2.5 V across it when fully charged.

A battery **Z** has an emf of 3.6 V and provides a current of 0.40 A for 2.5 hours .

0 6 . 3

Deduce whether **X** or **Z** stores more energy.

[2 marks]

0 6 . 4

A different supercapacitor **Y** can store the same amount of energy as the battery **Z**.

Suggest why it is better to use **Z** rather than **Y** to supply constant power to a component.

[1 mark]

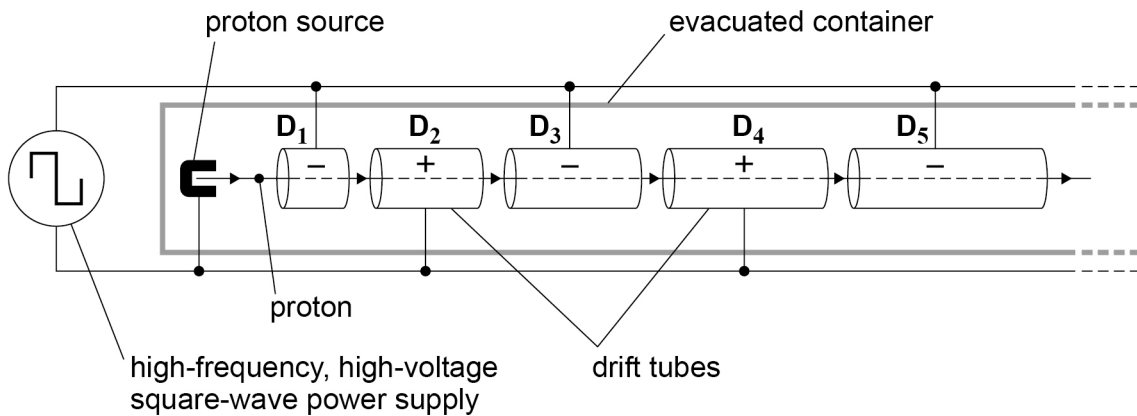


Figure 13 shows a linear accelerator that accelerates protons.

Protons pass through a series of hollow electrodes called drift tubes.
Drift tubes D_1 to D_5 are shown in **Figure 13**.

The drift tubes are connected to a high-frequency, high-voltage square-wave power supply so that adjacent drift tubes always have opposite polarities.
The periodic time of the supply is T .

Figure 13

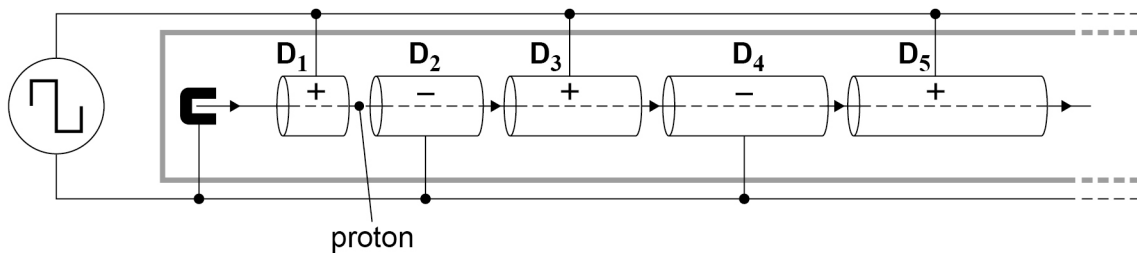


A proton with negligible energy is emitted from the proton source.

D_1 is negative and the proton accelerates towards it. The proton enters D_1 and travels at constant speed through it.

As the proton leaves D_1 , D_2 becomes negative and D_1 becomes positive. The proton accelerates towards D_2 , as shown in **Figure 14**.

Figure 14



The proton only accelerates in the gaps between the drift tubes and travels at constant speeds within the drift tubes.

0 7 . 1

Suggest why the proton travels at constant speed when inside any drift tube.

[1 mark]

When the proton crosses a gap, the potential difference between adjacent drift tubes is 42 kV.

The linear accelerator has 30 drift tubes D_1 to D_{30} .

0 7 . 2

Show that each proton has an energy of approximately 2×10^{-13} J when it reaches the end of the linear accelerator.

[2 marks]

Question 7 continues on the next page

Turn over ►



0 7 . 3

The frequency of the power supply is 5.0 MHz.

The proton spends a time $\frac{T}{2}$ in each drift tube.

Calculate the length of the final drift tube D_{30} .

[3 marks]

length = _____ m

0 7 . 4

A proton leaving the linear accelerator collides with an antiproton.

Describe the outcome of the collision, giving details of any products of the interaction.

rest energy of a proton = 1.51×10^{-10} J

[2 marks]

END OF SECTION A



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown. 

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown. 

You may do your working in the blank space around each question but this will not be marked. Do **not** use additional pages for this working.

0 8

An object performs simple harmonic motion described by:

$$x = A \cos \omega t$$

The velocity of the object is given by

[1 mark]

A $-A \sin \omega t$

☐

B $-\omega A \sin \omega t$

☐

C $A \sin \omega t$

☐

D $\omega A \sin \omega t$

☐

Turn over for the next question

Turn over ►



0 9

A mass is suspended from a spring.

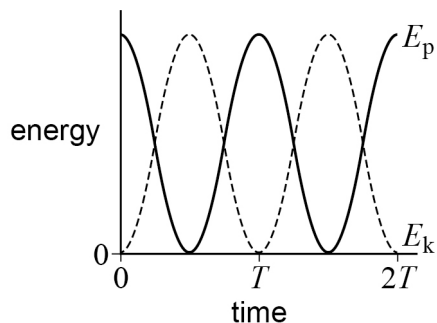
The mass is lifted a short distance above its equilibrium position and released at time $t = 0$

The mass performs simple harmonic motion with periodic time T .

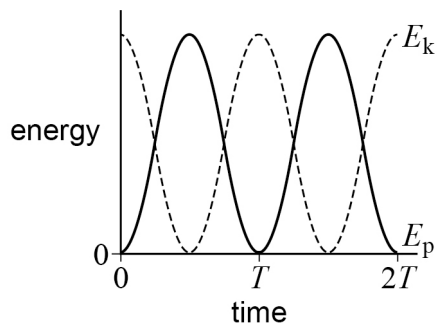
Which graph shows the variations with time of the kinetic energy E_k and the potential energy E_p of the mass–spring system?

[1 mark]

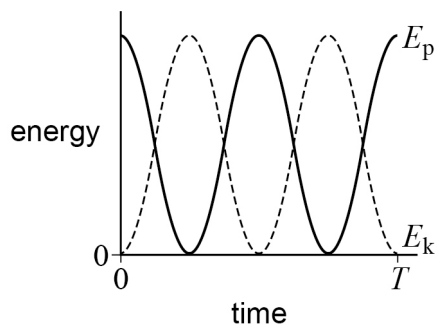
A



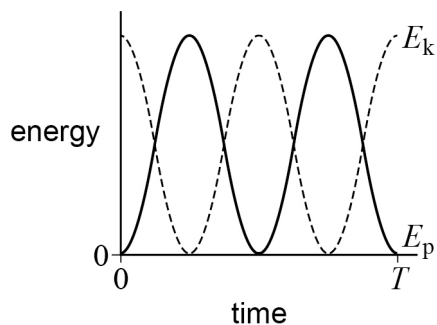
B



C



D



A



B



C



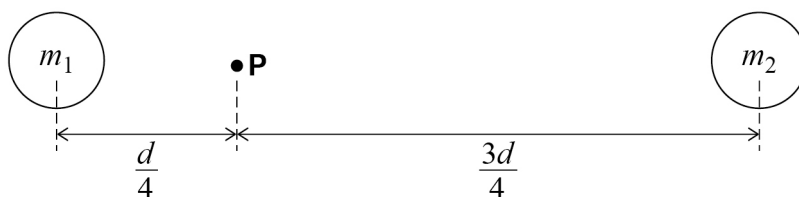
D



1 0

Two isolated masses m_1 and m_2 are separated by a distance d .

The gravitational field strength at point **P** is zero.



What is the relationship between m_1 and m_2 ?

[1 mark]

A $m_1 = \frac{m_2}{9}$ ☐

B $m_2 = \frac{m_1}{9}$ ☐

C $m_1 = \frac{m_2}{3}$ ☐

D $m_2 = \frac{m_1}{3}$ ☐

1 1

P is the gravitational force between two protons in a nucleus.

Q is the electrostatic force between two protons in a nucleus.

R is the gravitational force between an electron and the nucleus in an atom.

Which is correct?

[1 mark]

A $P < R < Q$ ☐

B $R < P < Q$ ☐

C $Q < R < P$ ☐

D $Q < P < R$ ☐



1 2

The electric field around a negative point charge is represented using lines of equipotential and electric field lines.

The potential difference between adjacent lines of equipotential is constant.

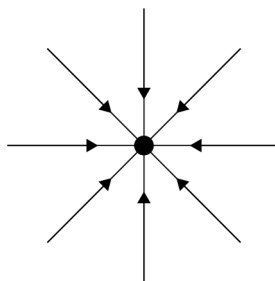
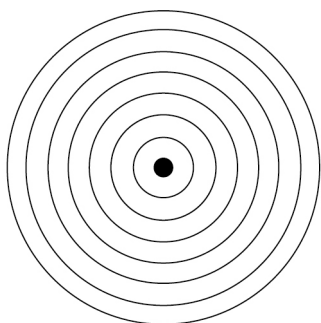
Which pair of drawings is correct?

[1 mark]

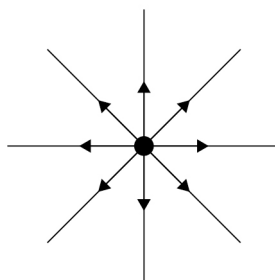
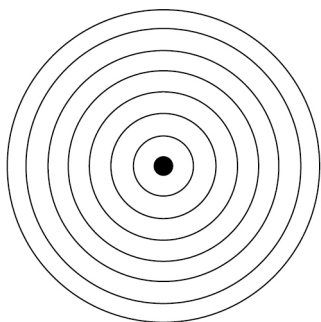
lines of equipotential

electric field lines

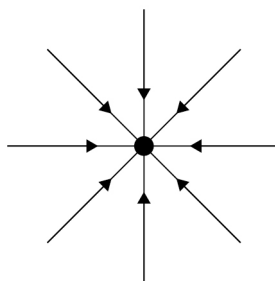
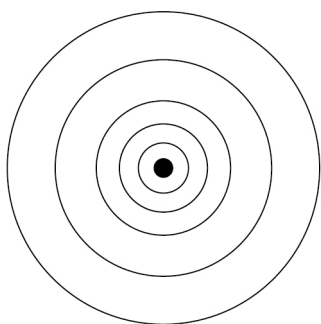
A



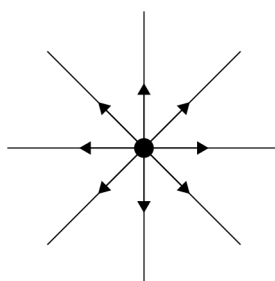
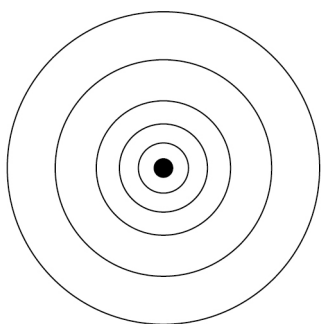
B



C



D



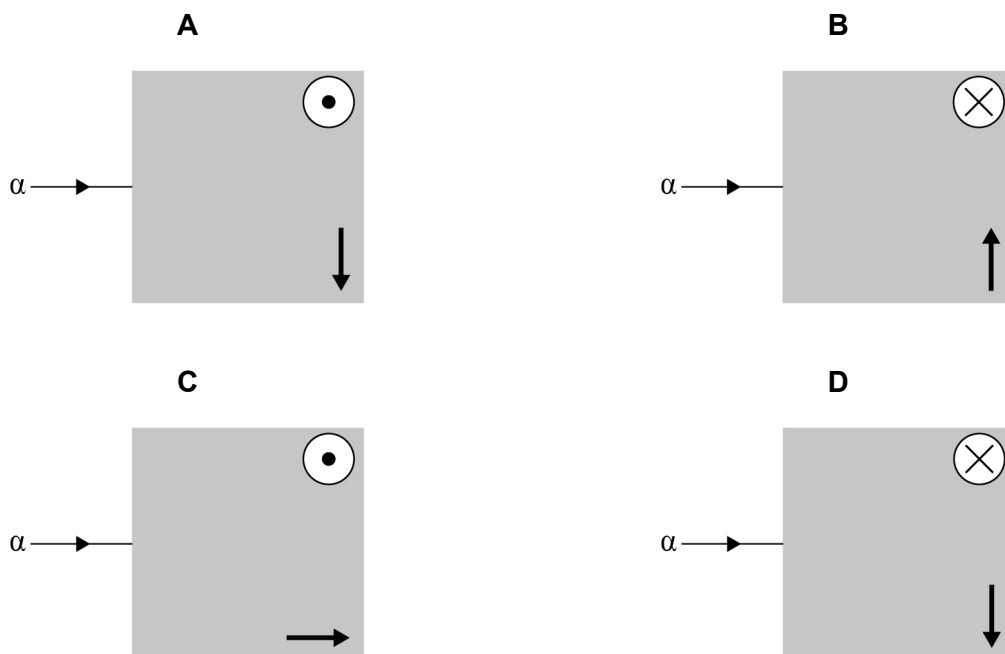
1 3 An alpha particle α enters a region where there is a uniform magnetic field and a uniform electric field.

The direction of the magnetic flux density is either into the page $\otimes$
or out of the page $\odot$

The direction of the electric field strength is given by one of the following arrows $\uparrow \downarrow \rightarrow$

Which diagram shows a combination of flux density and electric field strength that can result in the alpha particle travelling through this region at a constant velocity?

[1 mark]



- A** ☐
- B** ☐
- C** ☐
- D** ☐

Turn over for the next question

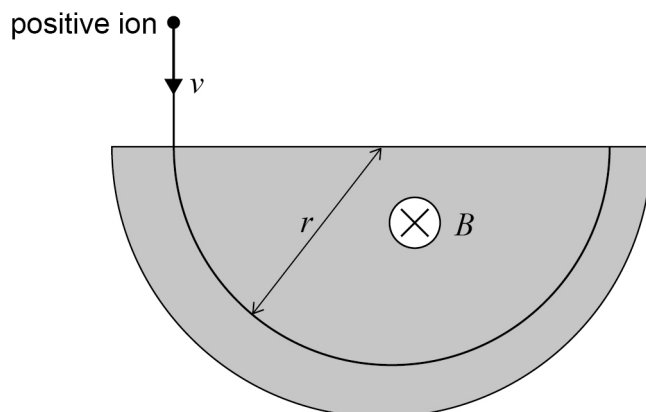
Turn over ►



1 4 Positive ions of mass m and charge q have a velocity v when they enter a uniform magnetic field.

The magnetic flux density B is into the page $\otimes$

The ions move along a semi-circular path of radius r .



The time taken for an ion to complete the semi-circular path depends only on:

[1 mark]

A q , B and m .

☐

B q , B and r .

☐

C q , m and r .

☐

D B , m and r .

☐

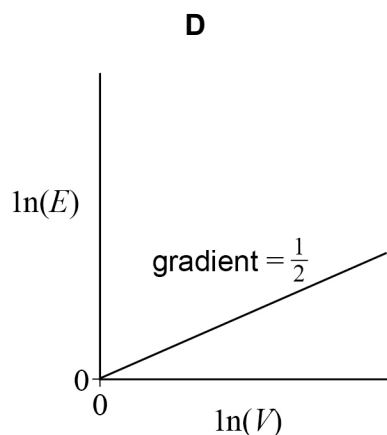
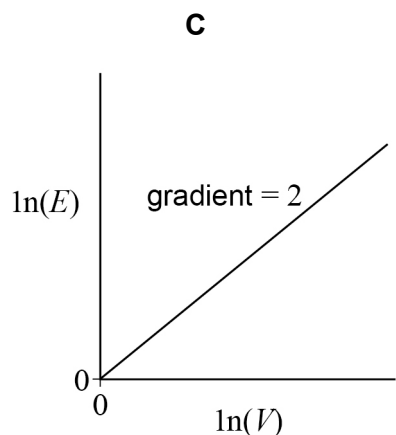
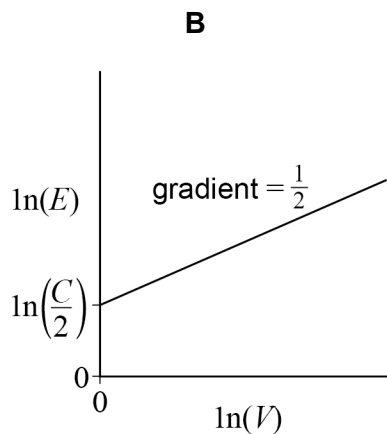
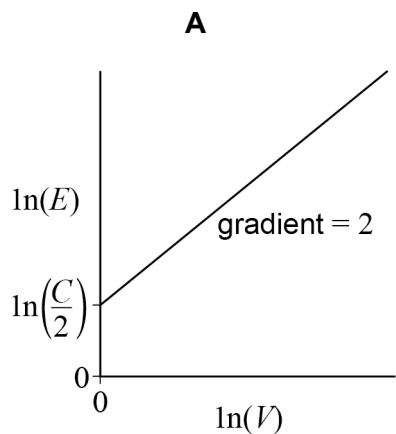
1 5

E is the energy stored on a capacitor of capacitance C .

E changes when the potential difference V across the capacitor changes.

Which graph shows the variation of $\ln(E)$ against $\ln(V)$ for the capacitor?

[1 mark]



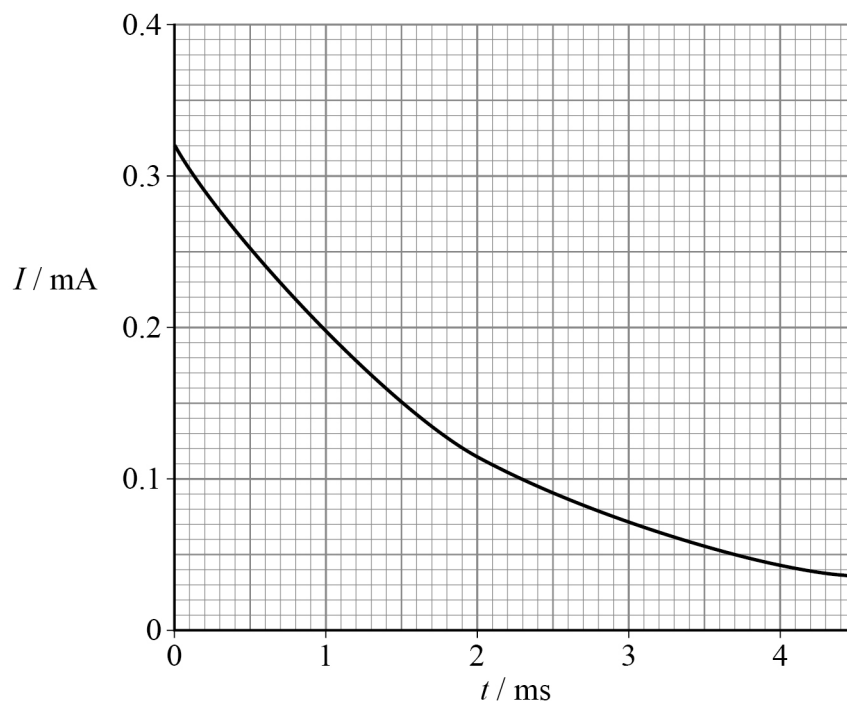
A ☐

B ☐

C ☐

D ☐

- 1 6** The graph shows the variation of current I with time t for a capacitor discharging through a $47\text{ k}\Omega$ resistor.



What is the capacitance of the capacitor?

[1 mark]

A 66 nF

☐

B 43 nF

☐

C 66 pF

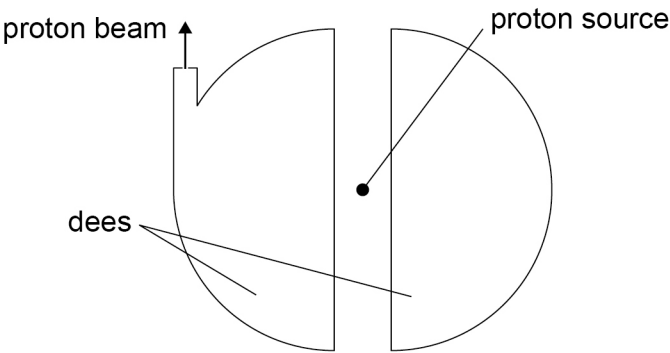
☐

D 43 pF

☐

1 7

The diagram shows a cyclotron.



Which row describes the velocity of the protons **inside** the dees and **between** the dees?
[1 mark]

	Velocity of protons inside the dees	Velocity of protons between the dees
A	changing	constant
B	changing	changing
C	constant	constant
D	constant	changing

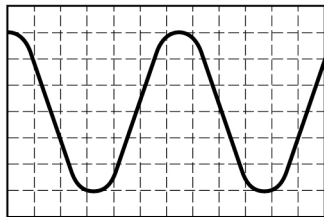
☐☐☐☐

Turn over for the next question

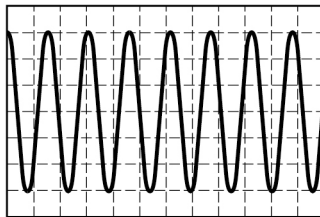


1 8

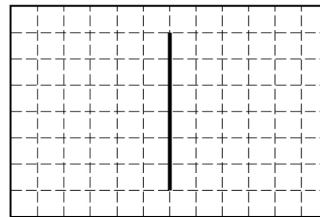
The diagrams show three oscilloscope traces of the same signal. The traces have different time-base settings.



1



2



3

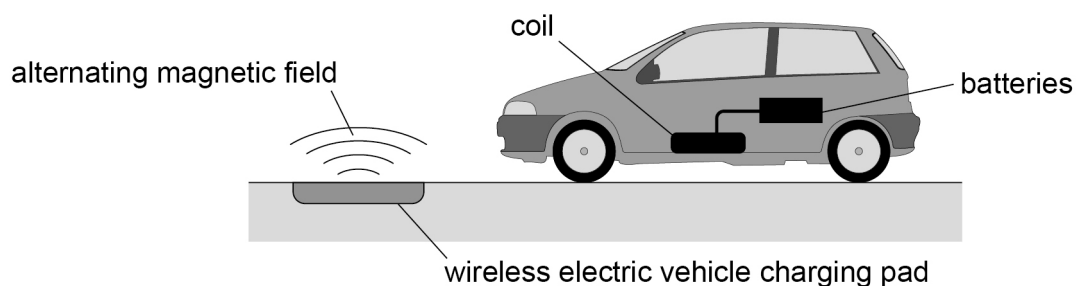
Which row shows the best alternative for determining the peak-to-peak voltage and the frequency of the signal?

[1 mark]

	Best trace for determining peak-to-peak voltage	Best trace for determining frequency
A	2	1
B	2	2
C	3	3
D	3	2

☐
☐
☐
☐


1 9 The diagram shows a system for recharging the batteries of moving electric vehicles.



The car passes through the alternating magnetic field.

An emf is induced in the coil, causing the car's batteries to charge.

Three suggestions for changes to the system are to:

- 1** increase the speed of the car
- 2** increase the area of the coil in the car
- 3** increase the frequency of the alternating magnetic field.

Which combination of suggestions will increase the energy supplied to the car battery as the car passes over the charging pad?

[1 mark]

- A** 1 and 2 only ☐
- B** 1 and 3 only ☐
- C** 2 and 3 only ☐
- D** 1 and 2 and 3 ☐

Turn over for the next question

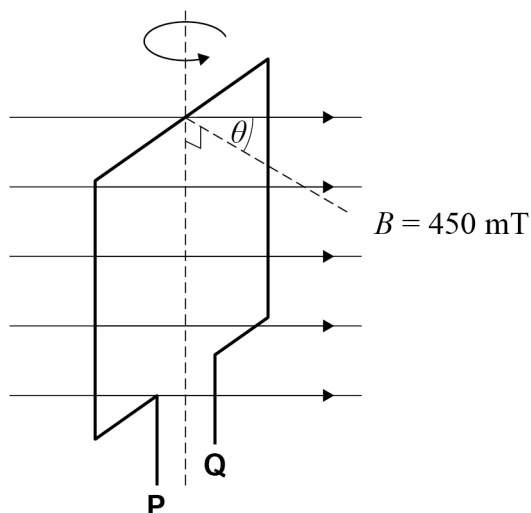
Turn over ►



2 0

A rectangular coil rotates at a constant angular speed about its axis in a uniform magnetic field of flux density 450 mT.

The coil is 5.0 cm long and 3.0 cm wide. It completes 200 rotations every minute.



At one instant, the angle θ between the magnetic field and the normal to the coil is 30° .

What is the emf induced between **P** and **Q** when θ is 30° ?

[1 mark]

A 120 V ☐

B 0.73 V ☐

C 0.012 V ☐

D 0.0071 V ☐



2 1 A power station produces 1.25 GW of electrical power at 275 kV rms. The transmission lines are 99% efficient.

Which row shows the resistance of the transmission lines and the rms voltage drop across them?

[1 mark]

	Resistance / Ω	rms voltage drop / V	
A	0.61	2750	☐
B	0.61	27 500	☐
C	6.1	2750	☐
D	6.1	27 500	☐

2 2 The power input to an ideal transformer is 7.2 MW at 132 kV rms. The rms voltage output from the transformer is 230 V.

What is the peak secondary current?

[1 mark]

- A** 390 A ☐
- B** 770 A ☐
- C** 22 kA ☐
- D** 44 kA ☐

[illegible]

This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

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0

1

2

Draw an arrow on **Figure 2** to show the direction of the resultant force on the person.

[1 mark]

Question 1 continues on the next page

Turn over ►



0 1 . 3

Explain how the station simulates the effect of the force of gravity on the person shown in **Figure 2**.

[2 marks]

0 1 . 4

The person has a linear momentum of 2500 kg m s^{-1} .

Calculate the mass of the person.

[3 marks]

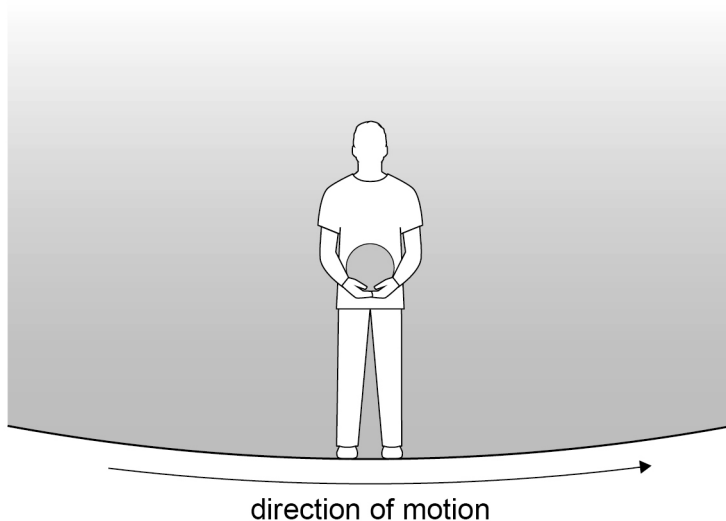
mass = _____ kg



0 1 . 5

Figure 3 shows the part of the space station where the person is standing. The person is holding a ball. The direction of motion of the station relative to an observer outside the station is shown.

Figure 3



The person releases the ball.

State and explain the motion of the ball, relative to the observer, after it is released.

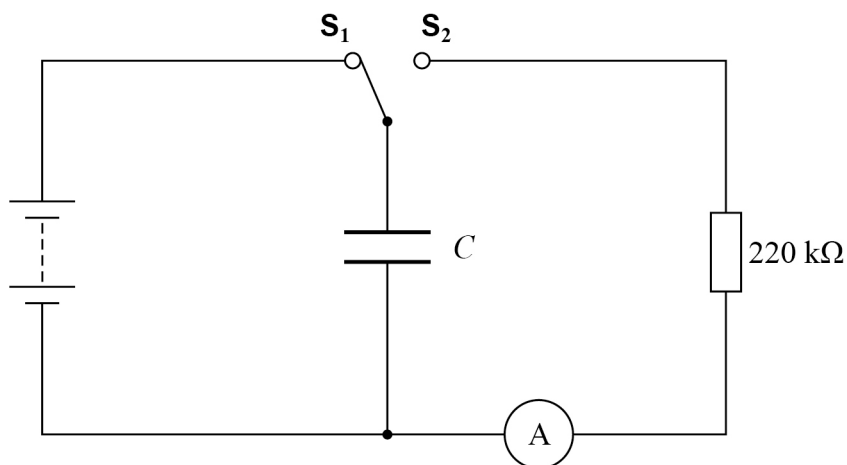
[3 marks]



Do not write

A student uses the circuit in **Figure 10** to determine the capacitance C of a capacitor. The resistance of the resistor is $220\text{ k}\Omega$.

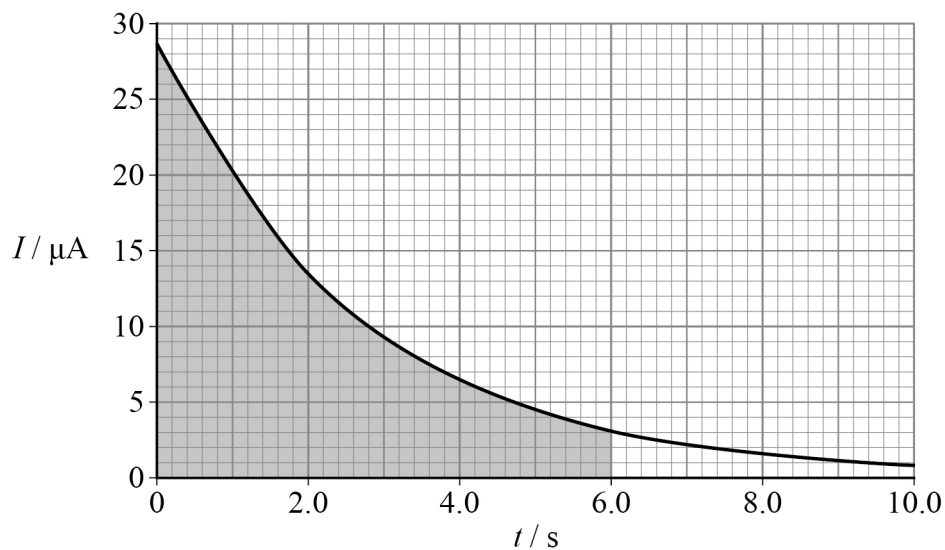
Figure 10



The switch is initially at S_1 and the capacitor is fully charged. At time $t = 0$, the switch is moved to S_2 and the capacitor discharges through the resistor.

Figure 11 shows the variation of the current I in the resistor with t .

Figure 11



0 5 . 1

Explain how the shaded area in **Figure 11** relates to the amount of charge that remains on the capacitor when $t = 6.0$ s.

[2 marks]

0 5 . 2

Determine the emf of the battery.

[2 marks]

emf = _____ V

0 5 . 3

Determine C .

[2 marks]

C = _____ F

Question 5 continues on the next page

Turn over ►



0 5 . 4 The student adapts the circuit in **Figure 10** using the same components.

This new circuit is used to measure the current in the $220\text{ k}\Omega$ resistor when:

- the capacitor is discharging through the resistor
- the capacitor is charging through the resistor.

Draw the circuit diagram for this circuit.

[2 marks]

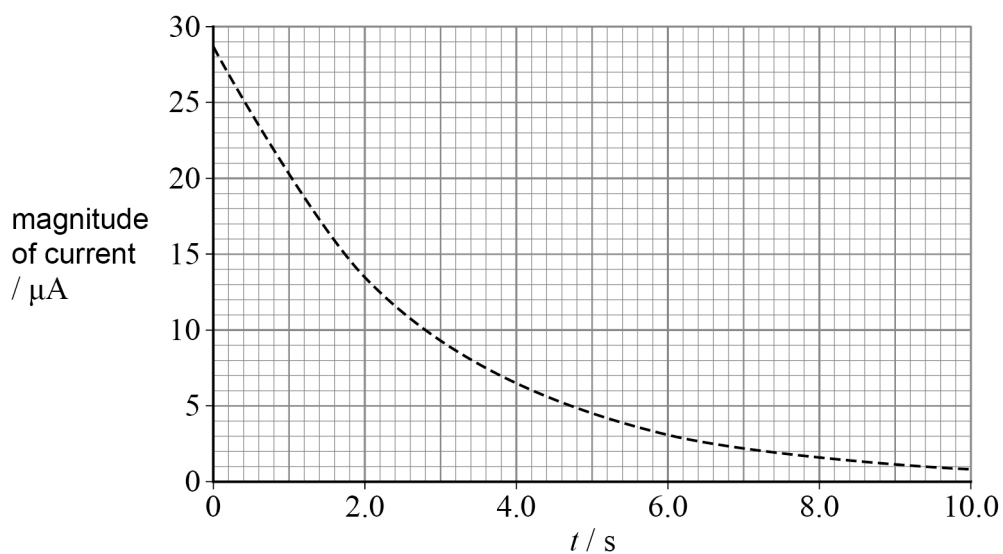
0 5 . 5 The capacitor is put in series with the battery and the $220\text{ k}\Omega$ resistor.
At time $t = 0$ the capacitor is uncharged.

Sketch, on **Figure 12**, the variation of the magnitude of the current with time as the capacitor charges.

The dashed line is the curve from **Figure 11**.

[2 marks]

Figure 12



Don't write

The distance between the Sun and the planet Mercury varies between 7.0×10^7 km and 4.6×10^7 km.

A force of gravity F acts on the Sun due to Mercury when the distance between their centres is 7.0×10^7 km.

What is the force of gravity acting on Mercury due to the Sun when the distance between their centres is 4.6×10^7 km?

[1 mark]

A $0.43F$

☐

B $0.66F$

☐

C $1.52F$

☐

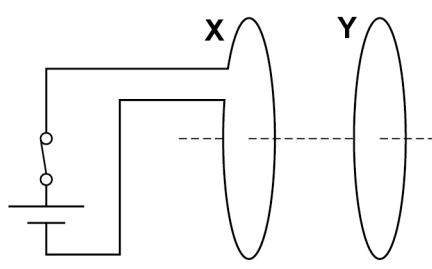
D $2.32F$

☐

1 2

A stationary circular loop of copper wire **X** is connected in series with a cell and a switch.

An identical stationary circular loop of copper wire **Y** is parallel to **X**. The centres of **X** and **Y** are on the same axis.



The switch is initially closed (on).

The switch is then opened (off).

Which row describes the force of **X** on **Y** immediately before and immediately after the switch is opened?

[1 mark]

	Immediately before	Immediately after	
A	there is a force	there is a force	☐
B	there is no force	there is a force	☐
C	there is a force	there is no force	☐
D	there is no force	there is no force	☐

Turn over for the next question



Do not write Do not write

1 4

A parallel-plate capacitor is made from two sheets of metal.

A dielectric of relative permittivity ϵ_r and thickness d fills the space between the sheets.

Which combination of ϵ_r and d gives the smallest capacitance?

[1 mark]

	ϵ_r	d / mm	
A	1.5	0.5	☐
B	2.0	0.7	☐
C	2.5	0.9	☐
D	3.2	1.1	☐

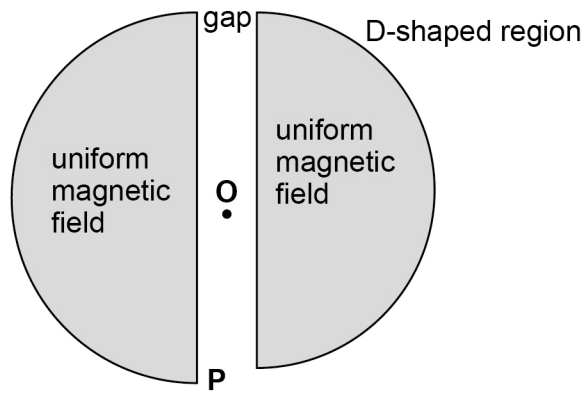
Turn over for the next question



Do not write

1 5

A proton released at **O** accelerates in a cyclotron and emerges at **P**.



Which row describes the path of the proton within a D-shaped region and in the gap?

[1 mark]

	Path in a D-shaped region	Path in the gap	
A	spiral	spiral	☐
B	spiral	straight	☐
C	circular	straight	☐
D	circular	circular	☐



Don't write

1 7

A capacitor is made from two parallel plates separated by an air gap. It is connected to a power supply and the capacitor stores a charge Q . The electric field strength between the capacitor's plates is E .

The capacitor is disconnected from the power supply. A dielectric is then inserted between the plates so that it completely fills the air gap.

Which row shows the charge on the capacitor and the electric field strength between the plates after the dielectric is inserted?

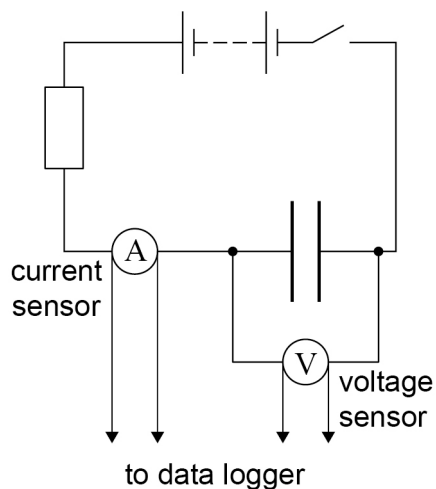
[1 mark]

	Charge	Electric field strength	
A	Q	$> E$	☐
B	Q	$< E$	☐
C	$< Q$	$> E$	☐
D	$< Q$	$< E$	☐

Figure 6 shows an arrangement for collecting data from a circuit that is used to charge a capacitor.

A data logger is connected to a current sensor and to a voltage sensor. The data are analysed.

Figure 6



The battery has an emf $\mathcal{E}$ and negligible internal resistance. The resistance of the resistor is R and the capacitance of the capacitor is C .

The capacitor is initially uncharged.

The switch is then closed and the voltage V_c across the capacitor and the current I are recorded. Further readings of V_c and I are recorded at regular time intervals.

It can be shown that:

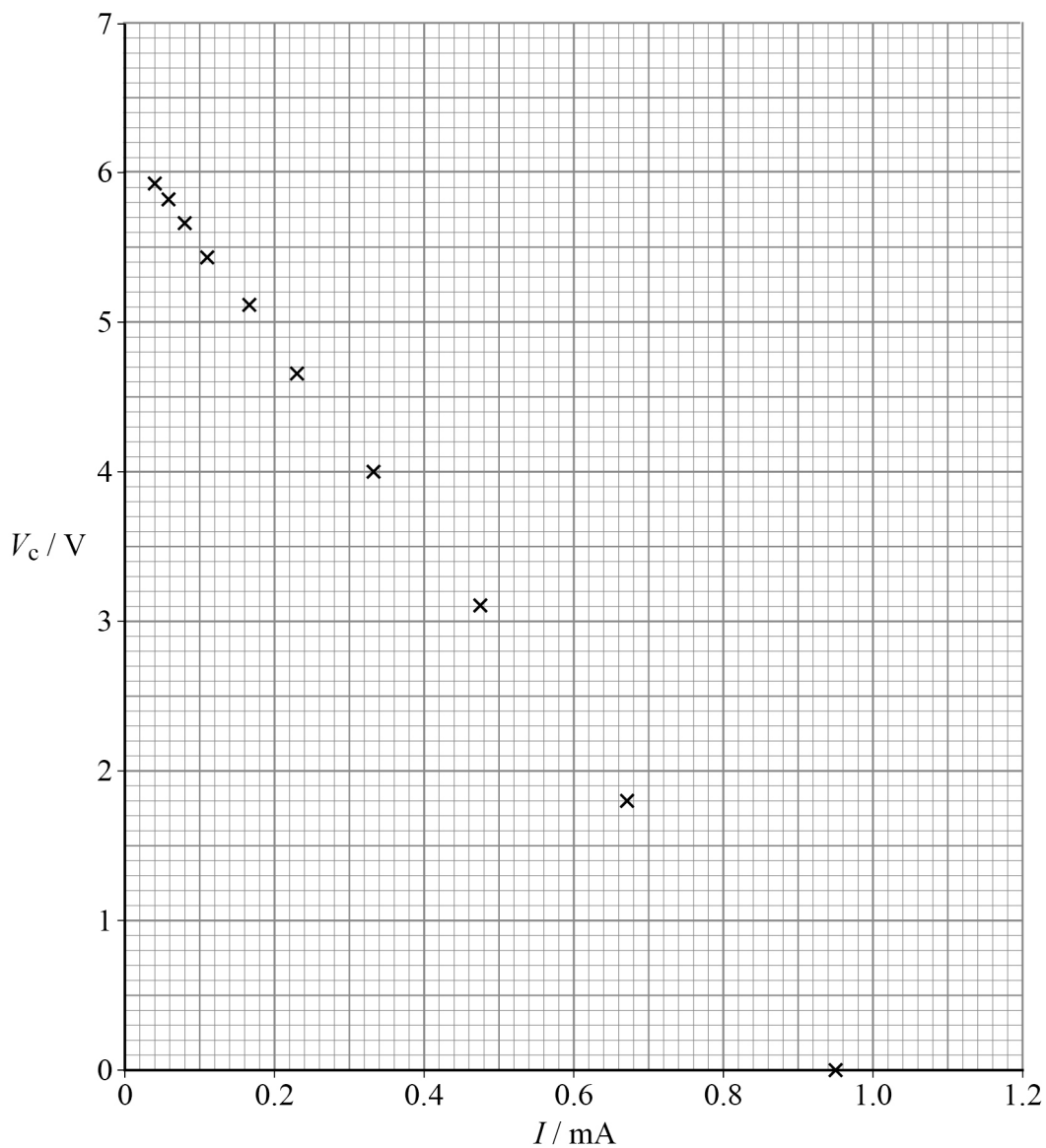
$$V_c = \mathcal{E} - IR$$

Question 4 continues on the next page



Figure 7 shows the plot of V_c against I .

Figure 7



Determine ε .

[1 mark]

$\varepsilon =$ _____ V

Determine R .

[2 marks]

$R =$ _____ Ω



The data shown on **Figure 7** were recorded at time intervals of 5.0 ms.

0 4 . 3 Explain why the plotted points on **Figure 7** become closer together as V_c increases.

[1 mark]

0 4 . 4 Determine the time constant for the circuit in **Figure 6** on page 13.
Go on to calculate C .

[5 marks]

time constant = _____ s

C = _____ F

Question 4 continues on the next page

Turn over ►

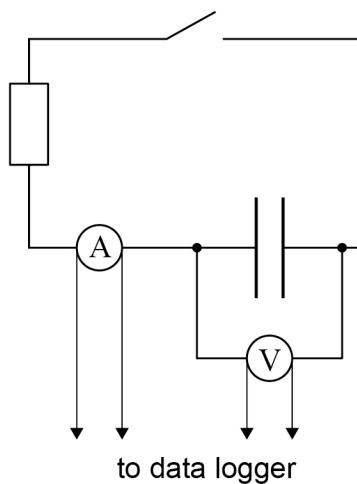


0 4 . 5

When the capacitor is fully charged the switch is opened. The battery is then removed from the circuit.

Figure 8 shows the new circuit for the charged capacitor.

Figure 8



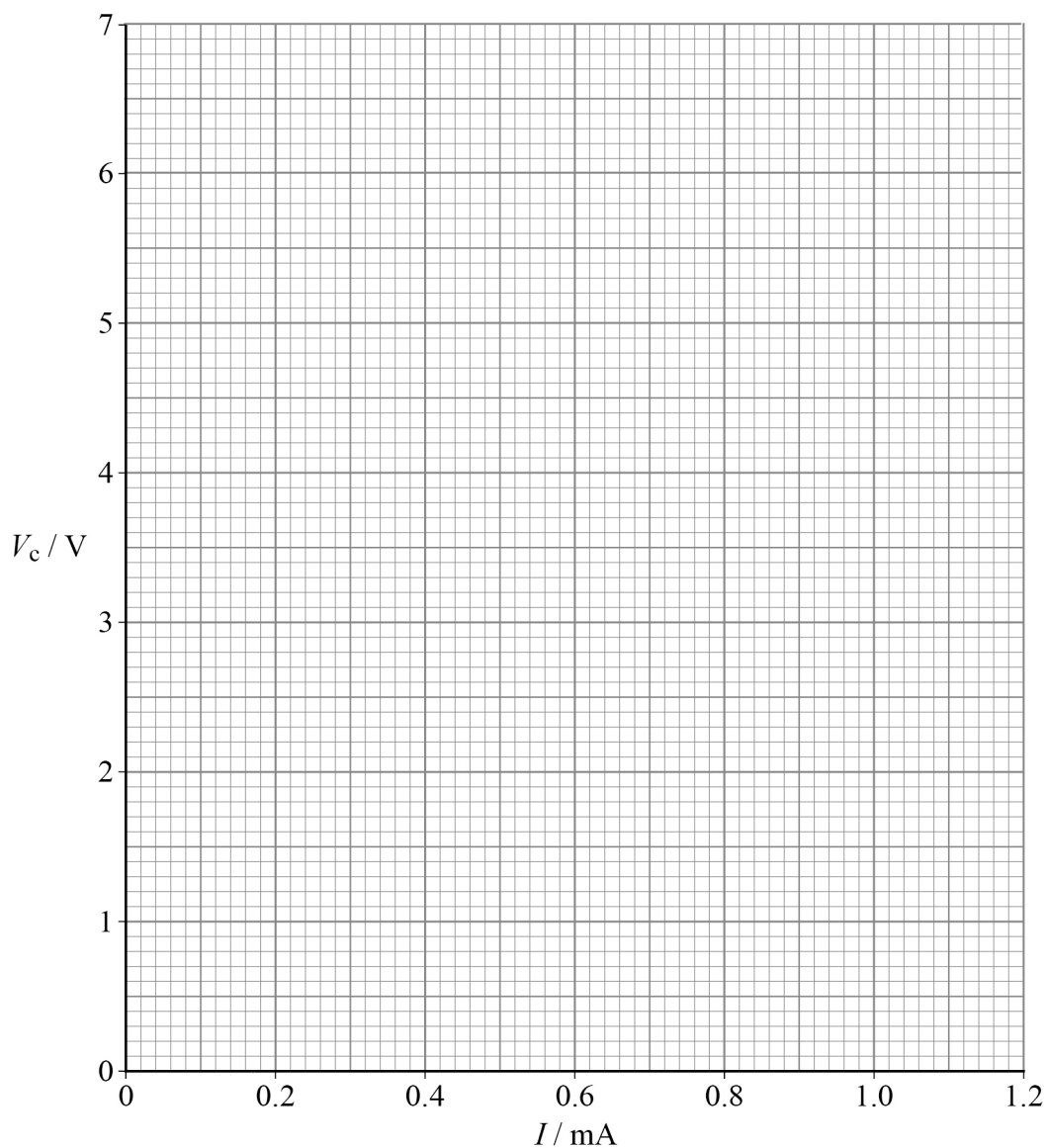
The switch is then closed and the capacitor discharges through the resistor.

Draw, on **Figure 9**, a line showing the variation of the magnitude of V_c with the magnitude of I for this experiment.

[2 marks]



Figure 9



0 4 . 6

Suggest why a data logger and sensors were used in these experiments rather than an ammeter, a voltmeter and a stopwatch.

[1 mark]



Do not write Do not write

0	5
---	---

Cobalt-60 (^{60}Co) is a radioactive isotope used in some school laboratories.

A school has a sealed ^{60}Co source with an activity of 62 kBq.

$$\begin{aligned}\text{half-life of } ^{60}\text{Co} &= 1.66 \times 10^8 \text{ s} \\ \text{molar mass of } ^{60}\text{Co} &= 60.0 \text{ g mol}^{-1}\end{aligned}$$

0	5	.	1
---	---	---	---

Calculate, in g, the mass of ^{60}Co in the sealed source.

[3 marks]

mass = _____ g

0	5	.	2
---	---	---	---

The school plans to replace the source when the activity of the ^{60}Co has decreased to 9.0 kBq.

Calculate, in years, the time before the source needs to be replaced.

[3 marks]

time = _____ years



0 6 . 1

Define the tesla.

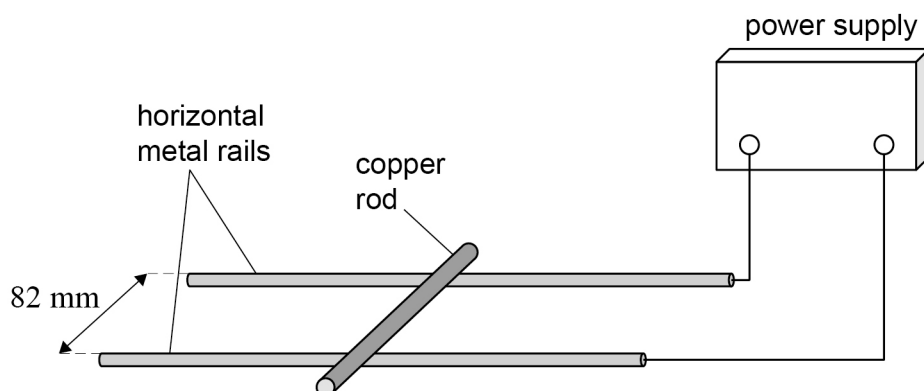
[1 mark]

For all parts of **Question 6**, assume that the Earth's magnetic flux density is negligible.

0 6 . 2

Figure 10 shows a dc power supply connected to two horizontal metal rails. A copper rod rests on the rails, completing the circuit. Initially the power supply is off.

Figure 10



A uniform magnetic field is applied vertically downwards between the two rails. When the power supply is switched on, the copper rod rolls to the left.

Label with a + sign the positive terminal of the power supply in **Figure 10**. Explain your answer.

[2 marks]

Question 6 continues on the next page

0 6 . 3

The power supply is switched off.

The rails are tilted at an angle of 10.0° to the horizontal.

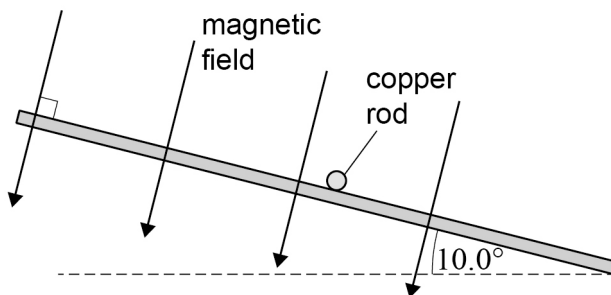
The direction of the magnetic field is changed so that it remains at 90° to the rails.

The power supply is switched on and the rod is placed on the rails. The combination of magnetic and gravitational forces acting on the rod keep it stationary.

Figure 11 shows a side view of the arrangement.

Figure 11

side view



The current in the rod is 2.75 A and the magnetic flux density is 0.11 T .

The distance between the rails is 82 mm .

Calculate the mass of the rod.

[3 marks]

mass = _____ kg



0 6 . 4

The connections to the power supply are removed.

The rails remain tilted at an angle of 10.0° to the horizontal and the magnetic field is unchanged.

The rod is placed on the rails and released.

The rod accelerates down the rails.

State and explain what happens to the emf induced across the rod.

[2 marks]

8

Turn over for the next question

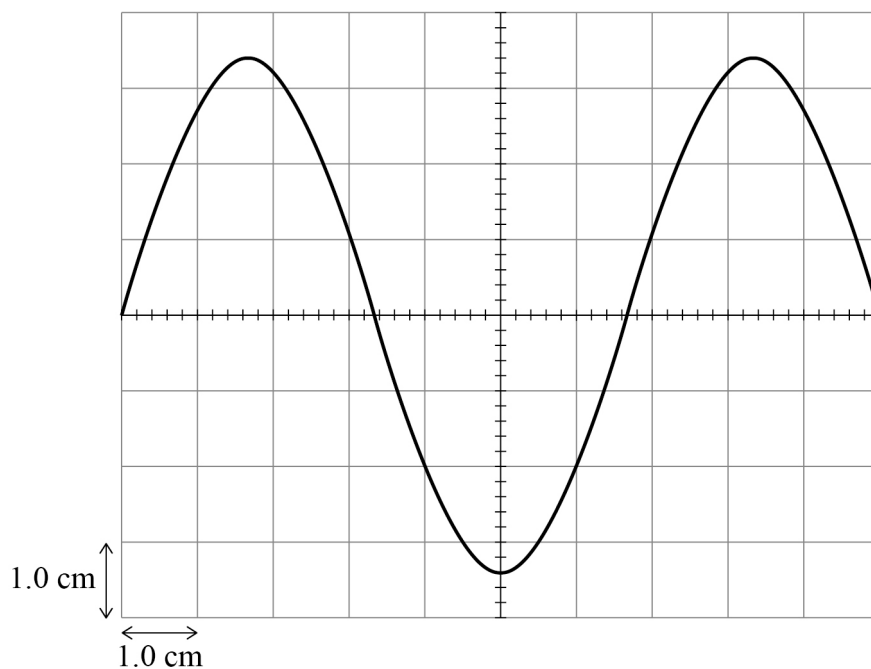
Turn over ►



0 7

Figure 12 shows the oscilloscope trace of the output of an ac power supply.

Figure 12



The root mean square (rms) voltage is 12.0 V and the frequency is 60.0 Hz.

0 7 . 1

Determine:

- the y -gain setting in V cm^{-1}
- the time-base setting in ms cm^{-1} .

[4 marks]

y -gain = _____ V cm^{-1}

time-base = _____ ms cm^{-1}



0 7 . 2

The output of the power supply is connected to the primary coil of a transformer.
The rms voltage output of the transformer is 3.0 V.
The number of turns on the secondary coil of the transformer is 4500

Calculate the number of turns on the primary coil.

[1 mark]

number of turns = _____

0 7 . 3

A resistor **R** is connected across the secondary coil.
The rms voltage across **R** is measured.

State the other measurements needed to determine the efficiency of the transformer.

[1 mark]

0 7 . 4

State **two** causes of inefficiency in a transformer.

[2 marks]

1 _____

2 _____

8

END OF SECTION A

Turn over ►



Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C** and **D**.

For each question select the best response.

Only **one** answer per question is allowed.

For each question, completely fill in the circle alongside the appropriate answer.

CORRECT METHOD



WRONG METHODS



If you want to change your answer you must cross out your original answer as shown.



If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown.



You may do your working in the blank space around each question but this will not be marked.
Do **not** use additional pages for this working.

0 8

A bob on the end of a light string is rotated in a vertical circle at a constant angular speed of 5.5 rad s^{-1} .

The bob has a mass of 0.20 kg and the circle has a radius of 0.75 m .

What is the **minimum** tension in the string?

[1 mark]

A 2.0 N

☐

B 2.6 N

☐

C 6.1 N

☐

D 6.5 N

☐


0 9

A car is travelling on a circular track.
The car completes four revolutions of the track in one minute.

What is the average angular speed of the car?

[1 mark]

A 0.030 rad s^{-1}

☐

B 0.10 rad s^{-1}

☐

C 0.42 rad s^{-1}

☐

D 1.6 rad s^{-1}

☐

1 0

A charged particle is moving with a velocity v at right angles to a magnetic field of flux density B .

The acceleration of the particle due to the magnetic field is a .

What is the specific charge of the particle?

[1 mark]

A $\frac{Bv^2}{a}$

☐

B $\frac{Bv}{a}$

☐

C $\frac{a}{Bv}$

☐

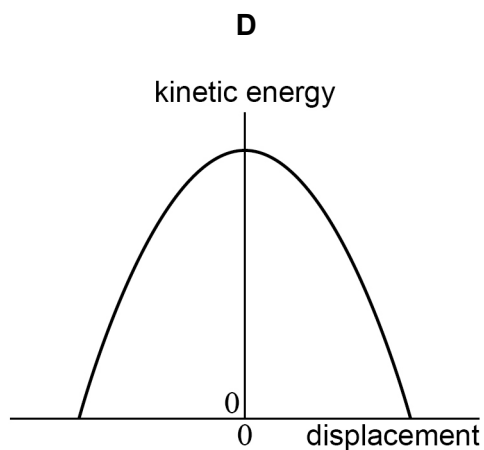
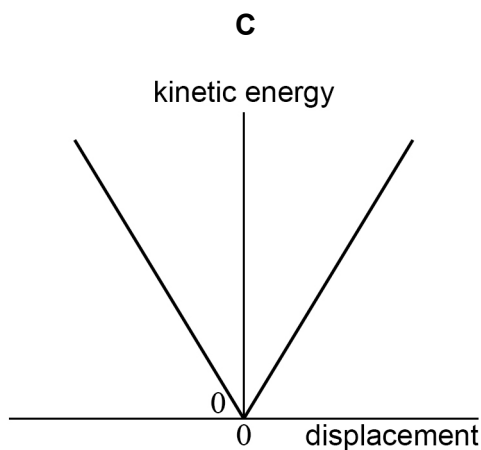
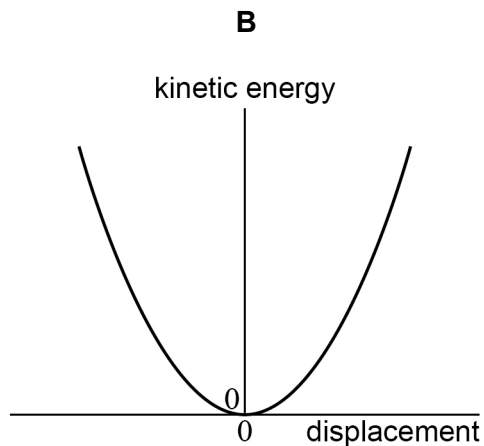
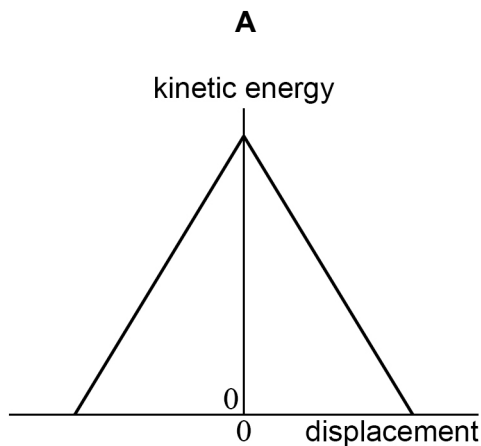
D $\frac{a}{Bv^2}$

☐


1 1 A mass on a spring undergoes simple harmonic motion.

Which graph shows the variation with displacement of the kinetic energy of the oscillating system?

[1 mark]



A ☐

B ☐

C ☐

D ☐



1 2

A satellite is in a circular orbit of radius R around a planet of mass M .
The orbital radius of the satellite is increased by a small distance Δr ,
where $\Delta r \ll R$.

The difference between the gravitational potentials of the two orbits is ΔV .

What is an approximation for ΔV ?

[1 mark]

A $\frac{GM\Delta r}{R^2}$ ☐

B $\frac{GM\Delta r}{R}$ ☐

C $\frac{GM}{R + \Delta r}$ ☐

D $\frac{GM}{\Delta r}$ ☐

1 3

A capacitor consists of two parallel plates with a dielectric between the plates.
The capacitor is connected to a battery and becomes fully charged.
The dielectric is then removed from between the plates without disconnecting the battery.

What happens to the charge on the plates and the energy stored on the capacitor?

[1 mark]

	Charge	Energy stored	
A	decreases	increases	☐
B	increases	decreases	☐
C	increases	increases	☐
D	decreases	decreases	☐



1 4

A charge of 1.5 mC is stored on a capacitor of capacitance 22 mF .
The capacitor is discharged until the pd across the capacitor is halved.

What is the change in the energy stored by the capacitor?

[1 mark]

A $13 \text{ } \mu\text{J}$ ☐

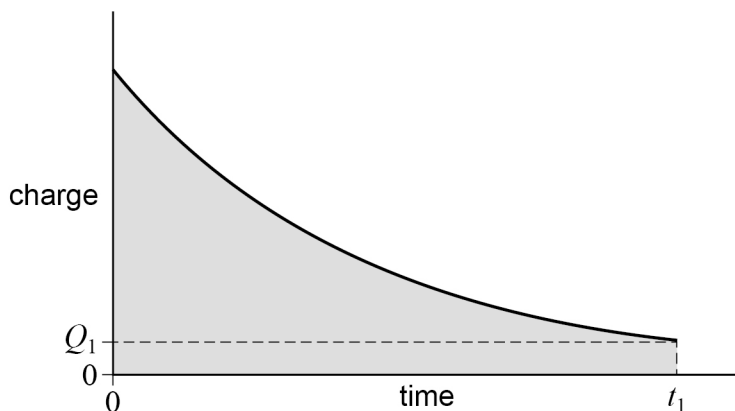
B $26 \text{ } \mu\text{J}$ ☐

C $38 \text{ } \mu\text{J}$ ☐

D $78 \text{ } \mu\text{J}$ ☐

1 5

A charged capacitor is discharged through a resistor of resistance R .
A graph is plotted of the charge on the capacitor against time.



After time t_1 the charge on the capacitor is Q_1 .

What is the capacitance of the capacitor?

[1 mark]

A $\frac{t_1}{R}$ ☐

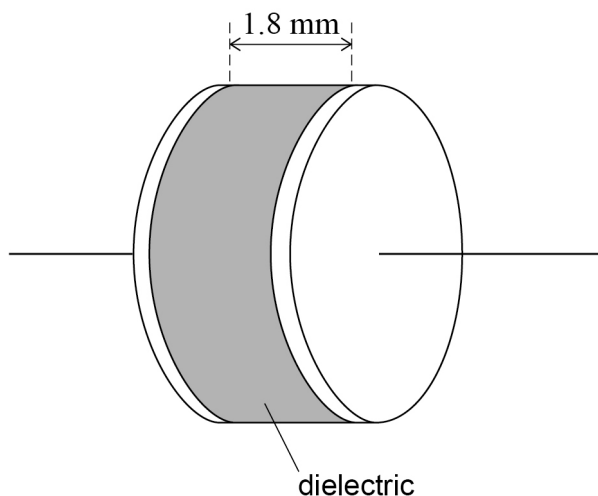
B $\frac{\text{area under the graph}}{RQ_1}$ ☐

C $-\frac{\text{area under the graph}}{R \times \text{gradient at } (t_1, Q_1) \times t_1}$ ☐

D $-\frac{Q_1}{R \times \text{gradient at } (t_1, Q_1)}$ ☐



1 6 A parallel-plate capacitor is made from two identical circular discs that completely overlap.



A sheet of dielectric material of thickness 1.8 mm fills the space between the plates.

The material has a relative permittivity of 2.4

The capacitance of the capacitor is 2.2 nF.

What is the diameter of the discs?

[1 mark]

A 0.24 m ☐

B 0.49 m ☐

C 0.58 m ☐

D 1.17 m ☐

Turn over for the next question

Turn over ►



1 7 At a distance y from an isolated point charge, the electric potential is V and the electric field strength is E .

What are the electric potential and the electric field strength at a distance $2y$ from the charge?

[1 mark]

	Electric potential	Electric field strength	
A	$\frac{V}{2}$	$\frac{E}{2}$	☐
B	$\frac{V}{2}$	$\frac{E}{4}$	☐
C	$\frac{V}{4}$	$\frac{E}{2}$	☐
D	$\frac{V}{4}$	$\frac{E}{4}$	☐

1 8 Which row identifies a vector quantity and a scalar quantity?

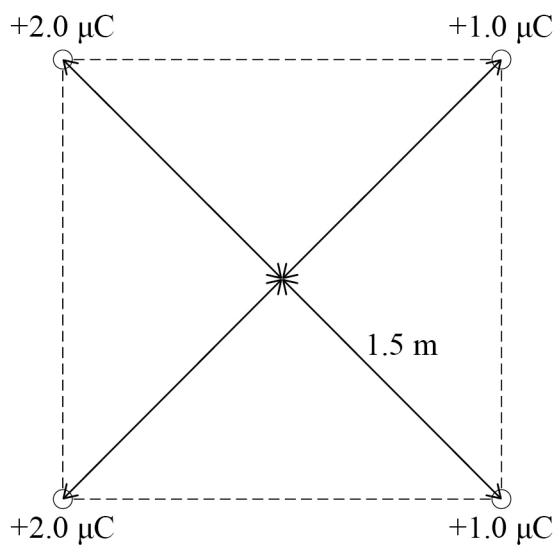
[1 mark]

	Vector	Scalar	
A	gravitational field strength	electric potential	☐
B	gravitational field strength	electric field strength	☐
C	gravitational potential	electric potential	☐
D	electric potential	gravitational field strength	☐



1 9

Four point charges are at the corners of a square.



What is the electric potential at the centre of the square?

[1 mark]

A $1.2 \times 10^4 \text{ V}$ ☐

B $2.4 \times 10^4 \text{ V}$ ☐

C $3.6 \times 10^4 \text{ V}$ ☐

D $4.0 \times 10^4 \text{ V}$ ☐

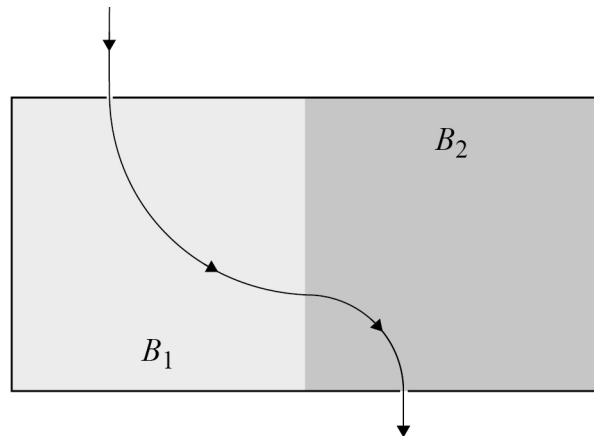
Turn over for the next question

Turn over ►



2 0

A proton passes through two evacuated chambers.
The chambers contain uniform magnetic fields of magnetic flux density B_1 and B_2 .
The path of the proton is at right angles to the magnetic field in each chamber.
The diagram shows the path of the proton from above.



$$x = \frac{\text{speed in second chamber}}{\text{speed in first chamber}}$$

Which row describes x and the directions of B_1 and B_2 ?

[1 mark]

	x	Direction of B_1	Direction of B_2	
A	>1	out of the page	into the page	☐
B	1	into the page	out of the page	☐
C	1	out of the page	into the page	☐
D	>1	into the page	out of the page	☐



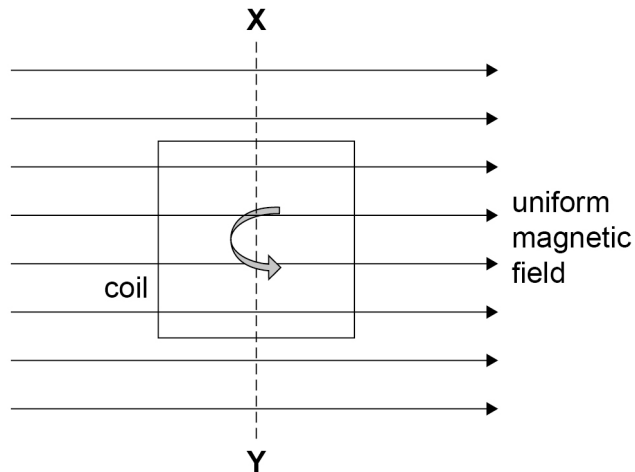
The following information is for Questions **21** and **22**.

A square coil of side 0.12 m has 20 turns.

The coil rotates with a frequency of 75 Hz in a uniform magnetic field.

The magnetic flux density is 45 mT.

The axis of rotation of the coil **XY** is at right angles to the magnetic field.



2 1 What is the maximum emf induced in the coil?

[1 mark]

A 0.30 V ☐

B 1.0 V ☐

C 6.1 V ☐

D 51 V ☐

2 2 What is the flux linkage at the instant when the minimum emf is induced in the coil?

[1 mark]

A 0 ☐

B 0.65 mWb ☐

C 13 mWb ☐

D 110 mWb ☐

[illegible]

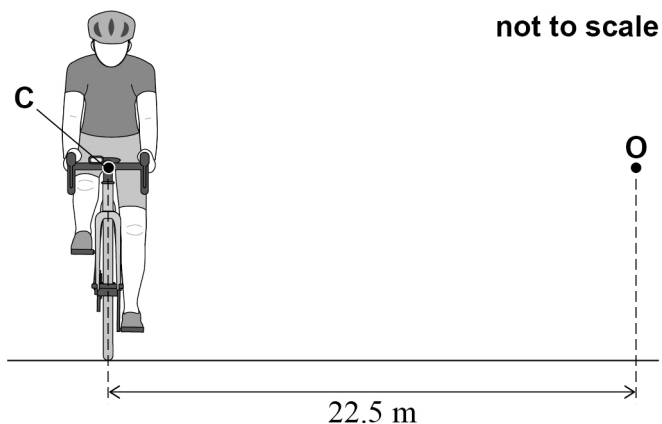
0 3

Figure 3 shows a person riding a bike on a horizontal track.

The bike and rider have a centre of mass at point **C**.

C moves in a circle about point **O**. The circle has a radius of 22.5 m.

Figure 3



The total mass of the bike and rider is 79.5 kg.

The bike and rider travel at a constant speed of 14.2 m s^{-1} .

The total frictional force acting on the bike is at its maximum value.

F_1 is the horizontal component of the frictional force acting towards **O**.

0 3 . 1

Show that F_1 is approximately 710 N.

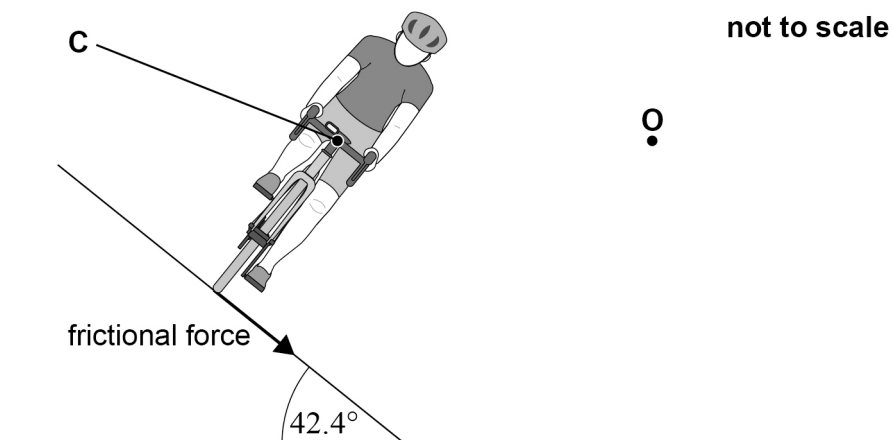
[1 mark]



A different cycling track is at an angle to the horizontal as shown in **Figure 4**.

The bike and rider travel in a horizontal circle about **O** on this track.

Figure 4



0 3 . 2

Annotate **Figure 4** with labelled arrows to show:

- the normal contact force **N** from the track on the bike
- the combined weight **W** of the bike and rider
- the resultant force **R**.

[2 marks]

Question 3 continues on the next page

The track in **Figure 4** is at an angle of 42.4° to the horizontal.

C moves around **O** in a horizontal circle of radius 22.5 m.

0 3 . 3 The bike and rider are travelling at a constant speed of 14.2 m s^{-1} and experience a normal contact force of 1040 N.

F_2 is the horizontal component of the frictional force acting on the bike.

Show that the magnitude of F_2 is less than the magnitude of F_1 .

[3 marks]



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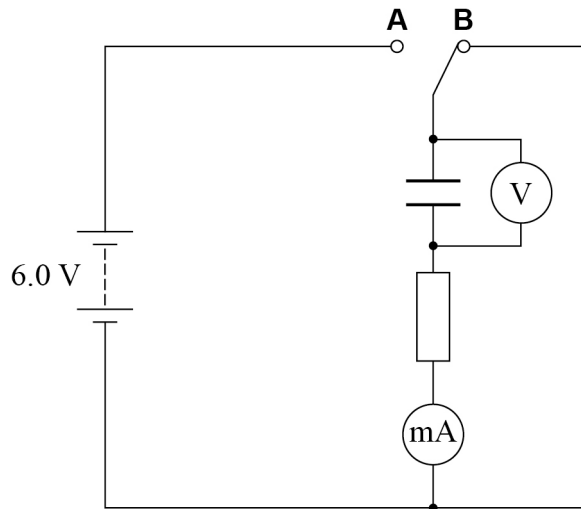
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Figure 9



The battery has an emf of 6.0 V and negligible internal resistance. Initially, the capacitor is fully discharged.

The switch is moved to position **A** at time $t = 0$

06.1 Calculate $\frac{\text{voltmeter reading}}{\text{emf of the battery}}$ after a time equal to one time constant.

[1 mark]

$\frac{\text{voltmeter reading}}{\text{emf of the battery}} =$ _____

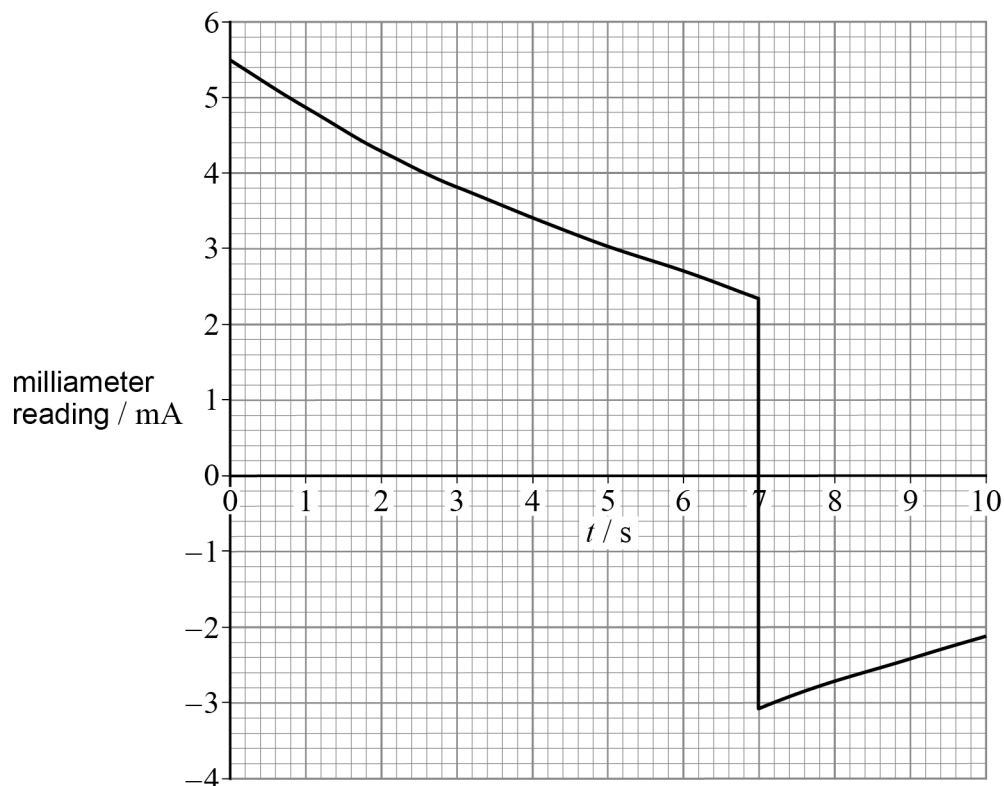
Question 6 continues on the next page



The switch is moved to position **B** at time $t = 7.0$ s.

Figure 10 shows the variation of the milliammeter reading with t .

Figure 10



0 6 . 2 Show that the resistance of the resistor is approximately 1000Ω .

[2 marks]



0 6 . 3

Determine the time constant and the capacitance of the capacitor in this circuit.

[3 marks]

time constant = _____ s

capacitance = _____ F

Question 6 continues on the next page



[6 marks]

[illegible]

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1**2**

Which is equivalent to the farad?

[1 mark]

A A s V^{-1}

☐

B $\text{A s}^{-1} \text{V}^{-1}$

☐

C C J^{-1}

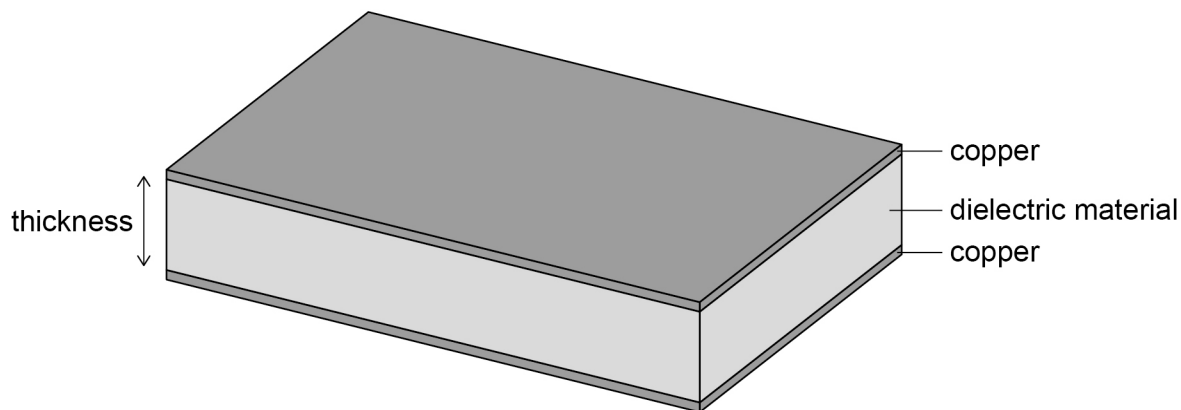
☐

D V C^{-1}

☐

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A student wants to construct a capacitor using two sheets of copper and a sheet of dielectric material. All of the sheets have the same area.



A sheet of dielectric material is inserted into the space between the sheets of copper so that the sheets of copper are in contact with the material.

The student has four sheets of dielectric material **A**, **B**, **C** and **D**, each with a different thickness and relative permittivity.

Which sheet produces a capacitor with the largest capacitance?

[1 mark]

	Thickness / mm	Relative permittivity
A	0.04	1.5
B	0.06	2.4
C	0.08	3.6
D	0.16	6.2



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04

An old design for a camera flash contains a capacitor and a filament lamp.
The capacitor supplies energy to the lamp for a short time to make the lamp flash.

The capacitor has a capacitance of $185 \mu\text{F}$.

The maximum charge stored by the capacitor is $3.7 \times 10^{-2} \text{ C}$.

04.1

Calculate the maximum energy stored by the capacitor.

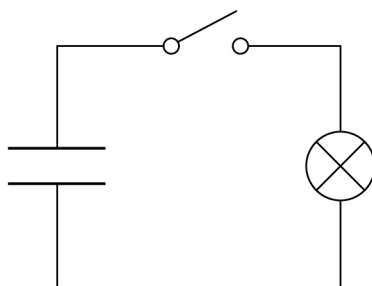
[1 mark]

maximum energy stored = _____ J

04.2

Figure 5 shows the charged capacitor in a circuit with the lamp and a switch.
At time $t = 0$, the switch is closed and the resistance of the lamp is 27Ω .

Figure 5



Calculate the initial current in the lamp when the switch closes.

[2 marks]

initial current = _____ A

Question 4 continues on the next page

Turn over ►

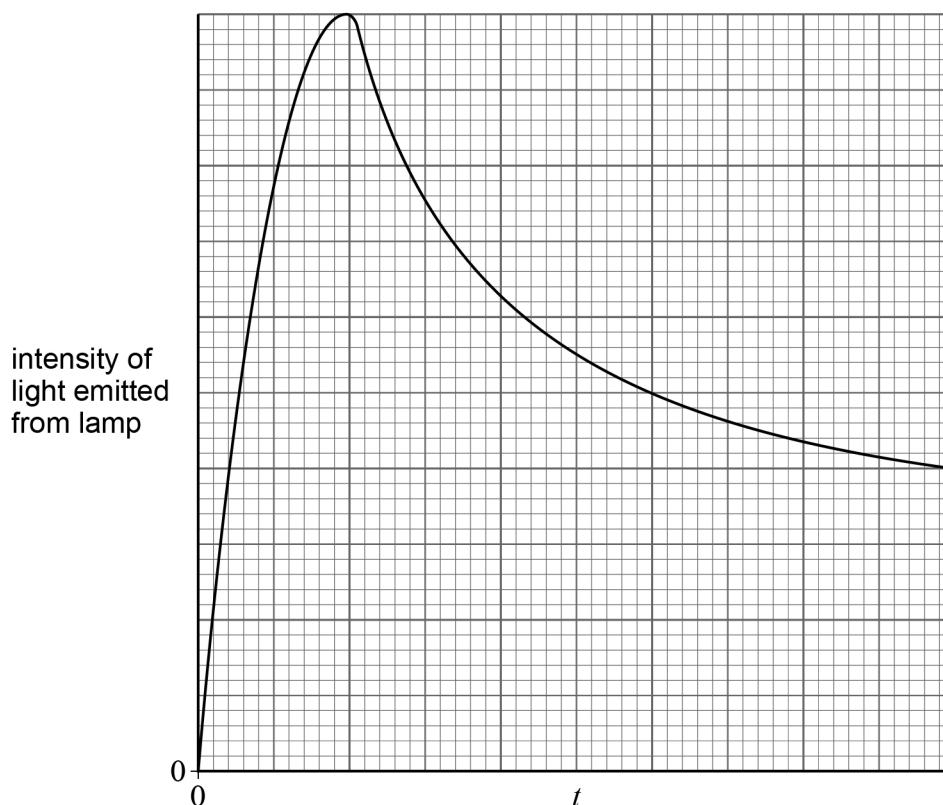


The filament in the lamp is made from metal.

The intensity of the light emitted by the lamp increases with the temperature of the filament.

Figure 6 shows how the intensity of the light varies with t during one flash.

Figure 6



0 4 . 3

Explain the variation of the intensity of the light emitted with t in **Figure 6**.

[4 marks]



0 4 . 4

The intensity of the light emitted by the lamp reaches a maximum value shortly after the switch is closed.

Determine whether the intensity of the light emitted by the lamp decreases exponentially with time after it reaches this maximum value.

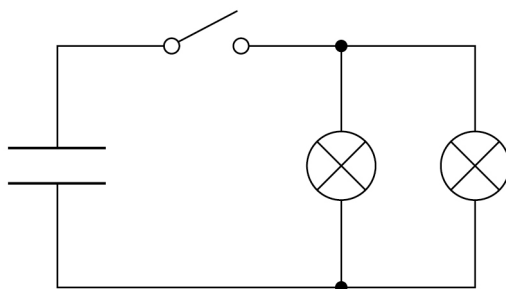
[2 marks]

0 4 . 5

The capacitor is fully charged.

A second identical lamp is added in parallel with the first lamp as shown in **Figure 7**.

Figure 7



A student suggests that the addition of the second lamp will increase the duration of one flash.

Explain whether the student is correct.

[1 mark]



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A capacitor is discharged through a resistor and loses 50% of its charge in a time of 10 s.

What is the time constant of the circuit?

[1 mark]

A 5 s

☐

B 7 s

☐

C 10 s

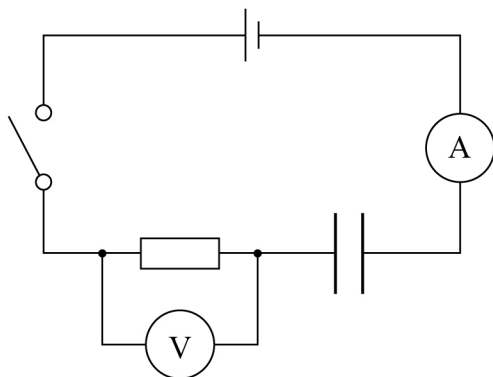
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D 14 s

☐

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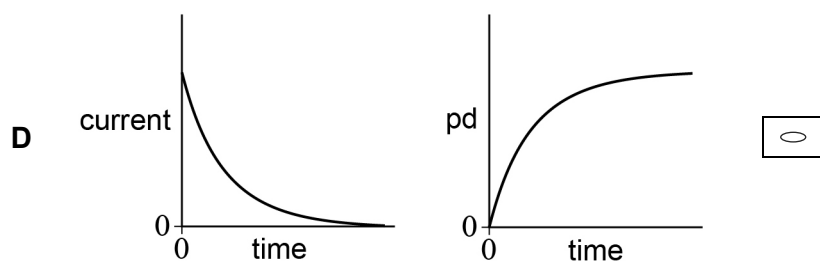
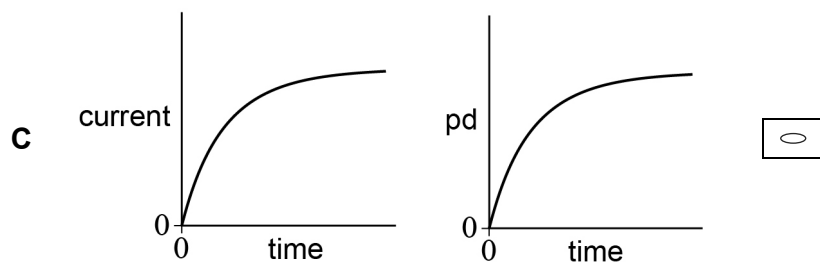
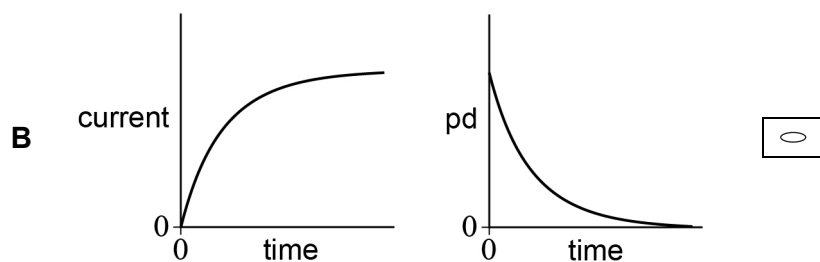
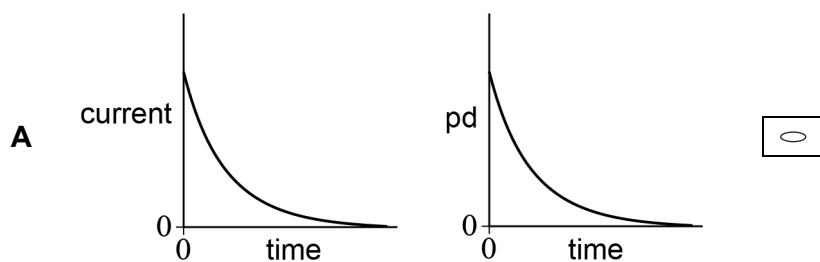
The circuit shown is used to charge an uncharged capacitor.



The switch is closed at a time equal to zero.

Which pair of graphs shows the variation of current with time and the variation of potential difference (pd) with time?

[1 mark]



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1 6

An uncharged capacitor of capacitance C is fully charged by a cell of emf ε and negligible internal resistance, through a resistor of resistance R .

What is the total energy dissipated by the resistor?

[1 mark]

A $\frac{1}{2} C \varepsilon^2$

☐

B $C \varepsilon^2$

☐

C $\frac{\varepsilon^2}{R}$

☐

D $\frac{1}{2} \frac{\varepsilon^2}{R}$

☐

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0	2
---	---

An ancient sealed flask contains a liquid, assumed to be water. An archaeologist asks a scientist to determine the volume of liquid in the flask without opening the flask. The scientist decides to use a radioactive isotope of sodium ($^{24}_{11}\text{Na}$) that decays with a half-life of 14.8 h.

0	2
---	---

1

The scientist first mixes a compound that contains 3.0×10^{-10} g of sodium-24 with 1500 cm^3 of water. The scientist then injects 15 cm^3 of the solution into the flask through the seal.

Show that initially about 7.5×10^{10} atoms of sodium-24 are injected into the flask.

[1 mark]

0	2
---	---

2

Show that the initial activity of the solution that is injected into the flask is about $1 \times 10^6 \text{ Bq}$.

[3 marks]

activity = _____ Bq

0 2 . 3

The scientist waits for 3.5 h to allow the injected solution to mix thoroughly with the liquid in the flask. The scientist then extracts 15 cm^3 of the liquid from the flask and measures its activity which is found to be 3600 Bq.

Calculate the total activity of the sodium-24 in the flask after 3.5 h and hence determine the volume of liquid in the flask.

[3 marks]

total activity = _____ Bq

volume of liquid = _____ m^3

0 2 . 4

The archaeologist obtained an estimate of the volume knowing that similar empty flasks have an average mass of 1.5 kg and that mass of the flask and liquid was 5.2 kg.

Compare the estimate that the archaeologist could obtain from these masses with the volume calculated in Question **02.3** and account for any difference.

[2 marks]

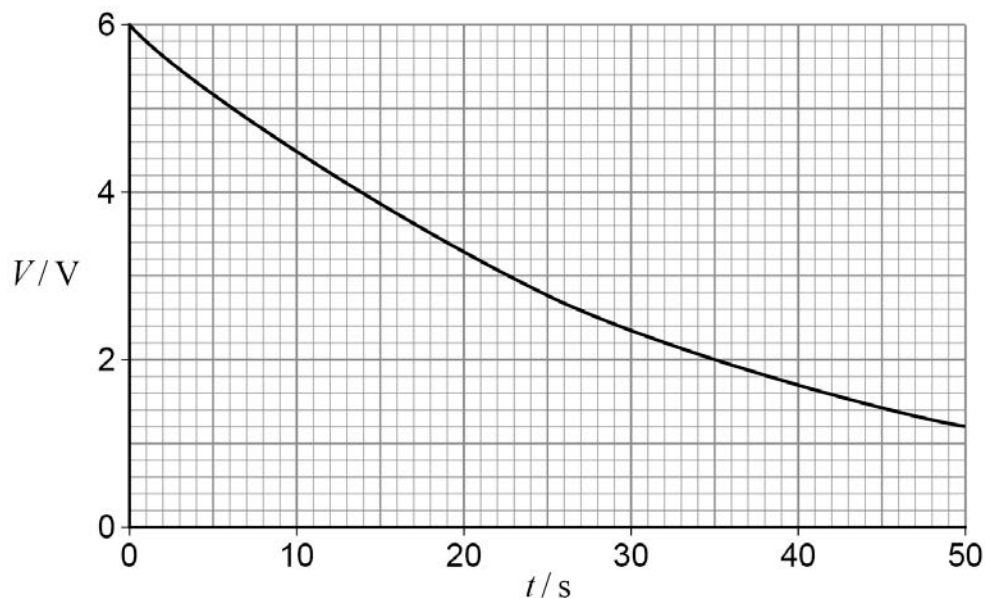
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0 7 . 0

A charged $4.7 \mu\text{F}$ capacitor initially stores $85 \mu\text{J}$ of energy. The capacitor is discharged through a resistor **R**.

Figure Z shows the variation of potential difference V across the capacitor with discharge time t .

Figure Z



0 7 . 1

Calculate the charge on the capacitor when $t = 0$

[1 mark]

charge = _____ C

0 7 . 2

Calculate the energy transferred to **R** when the capacitor has been discharging for 50 s.

[3 marks]

energy transferred to **R** = _____ J

0 7 . 3 Calculate the resistance of **R**.

[4 marks]

resistance= _____ Ω

Turn over for the next question

Section B

Each of the questions in this section is followed by four responses, **A**, **B**, **C**, and **D**. For each question select the best response.

Only **one** answer per question is allowed.

For each answer completely fill in the circle alongside the appropriate answer.



If you want to change your answer you must cross out your original answer as shown. 

If you wish to return to an answer previously crossed out, ring the answer you now wish to select as shown. 